



Robust mixture regression modeling based on the normal mean-variance mixture distributions

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ABSTRACT

Mixture regression models (MRMs) are widely used to capture the heterogeneity of relationships between the response variable and one or more predictors coming from several non-homogeneous groups. Since the conventional MRMs are quite sensitive to departures from normality caused by extra skewness and possible heavy tails, various extensions built on more flexible distributions have been put forward in the last decade. The class of normal mean-variance mixture (NMVM) distributions that arise from scaling both the mean and variance of a normal random variable with a common mixing distribution encompasses many prominent (symmetric or asymmetrical) distributions as special cases. A unified approach to robustifying MRMs is proposed by considering the class of NMVM distributions for component errors. An expectation conditional maximization either (ECME) algorithm, which incorporates membership indicators and the latent scaling variables as the missing data, is developed for carrying out maximum likelihood (ML) estimation of model parameters. Four simulation studies are conducted to examine the finite-sample property of ML estimators and the robustness of the proposed model against outliers for contaminated and noisy data. The usefulness and superiority of our methodology are demonstrated through applications to two real datasets.

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1. Introduction

Mixture regression models (MRMs), which initially emerged as ‘switching regression’ in the economics literature (Quandt, 1972; Goldfeld and Quandt, 1973), have been widely used to explore the source of heterogeneity when groups of individuals respond to a predictor differentially. Alternatively, Späth (1979) gave the name of ‘clusterwise linear regression’ and offered an ordinary least squares estimation method to minimize the sum of squared errors within each group. Over the past few decades, the MRM has been used in a large number of applications in a broad range of scientific disciplines, see Liu and Lin (2014) for a comprehensive overview.

Let $\mathbf{x}_j = (1, x_{1j}, \dots, x_{pj})^\top$ be a $(p + 1) \times 1$ vector of fixed explanatory variables corresponding to a scalar response variable Y_j , for $j = 1, \dots, n$. Moreover, we introduce vectors of unobservable membership indicators $\mathbf{Z}_j = (Z_{1j}, \dots, Z_{gj})^\top$ with entry Z_{ij} which equals one if Y_j comes from the i th group ($i = 1, \dots, g$) and zero otherwise. The mixing probabilities $\Pr(Z_{ij} = 1) = \pi_i$, for $i = 1, \dots, g$, are subject to the constraint of $\sum_{i=1}^g \pi_i = 1$, where g is the number of components in

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the MRM. Thus, the random vector $\mathbf{Z}_j$ follows a multinomial distribution with one trial and cell probabilities $(\pi_1, \dots, \pi_g)$, denoted by $\mathbf{Z}_j \sim \mathcal{M}(1; \pi_1, \dots, \pi_g)$. The classical MRM, named as N-MRM henceforth, can be formulated as follows:

$$Y_j = \mathbf{x}_j^\top \boldsymbol{\beta}_i + \epsilon_{ij}, \quad \text{with mixing probability } \pi_i, \quad (1)$$

where $\boldsymbol{\beta}_i = (\beta_{0i}, \beta_{1i}, \dots, \beta_{pi})^\top$ is a $(p+1) \times 1$ vector of regression coefficients for component i with intercept β_{0i} , and $\epsilon_{ij} \sim N(0, \sigma_i^2)$ are random errors assumed to be mutually independent. For model (1), denote by $\boldsymbol{\Theta} = \{\boldsymbol{\pi}, \boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_g\}$ the collection of all unknown parameters to be estimated, where $\boldsymbol{\pi} = (\pi_1, \dots, \pi_{g-1})$ and $\boldsymbol{\theta}_i = \{\boldsymbol{\beta}_i, \sigma_i^2\}$. The ML estimate of $\boldsymbol{\Theta}$ is traditionally obtained by searching the solution of the following nonlinear convex function:

$$\hat{\boldsymbol{\Theta}} = \arg \max_{\boldsymbol{\Theta}} \sum_{j=1}^n \log \left(\sum_{i=1}^g \pi_i \phi(y_j; \mathbf{x}_j^\top \boldsymbol{\beta}_i, \sigma_i^2) \right), \quad (2)$$

where $\phi(\cdot; \mu, \sigma^2)$ stands for the pdf of normal distribution with mean μ and variance σ^2 . However, the task of (2) is intractable since the log-likelihood function is stated as a sum of natural logarithms rather than a logarithm of a sum. As such, the derivatives of the log-likelihood involve ratios of a weighted sum of normal densities, which are intractable to solve the roots explicitly. Therefore, one usually resorts to more elaborate techniques such as the expectation maximization (EM) algorithm (Dempster et al., 1977) for finding $\hat{\boldsymbol{\Theta}}$ within a complete-data framework (Aitkin and Wilson, 1980). An alternative way of estimating parameters in (1) is to exploit an EM-like algorithm for the semi-parametric mixture model introduced by Benaglia et al. (2009). The use of semi-parametric approach could be much more flexible than other existing algorithms, and might be applicable even for the non-identifiable model. Generally, model (1) is identifiable under the conditions provided by Hennig (2000) and Frühwirth-Schnatter (2006, Section 8.2.2). These conditions are usually satisfied when the domain of covariates contains an open set in $\mathbb{R}^p$. To overcome the non-identifiability due to label switching (Stephens, 2000), we adopt a constraint on the mixing proportions $\pi_1 < \dots < \pi_g$ (Richardson and Green, 1997), for relabeling model parameters.

Although the N-MRM is predominant in the literature due to its mathematical convenience and simplicity, the normality assumption is often violated in many practical situations, especially when the data contain outliers or highly asymmetric observations. To deal with departures from normality, a great deal of various robust estimation methods and modeling procedures have been proposed. The recent burgeoning of non-normal extensions of MRM include the Student's t distribution (Yao et al., 2014; Galimberti and Soffritti, 2014), the Laplace distribution (Song et al., 2014), the contaminated normal distribution (Punzo and McNicholas, 2017), the skew-normal distribution (Liu and Lin, 2014), and the scale mixtures of skew-normal (SMSN) distributions (Zeller et al., 2016; Benites et al., 2019), among others.

This paper aims at enriching the palette of non-normal MRM by considering the class of normal mean-variance mixture (NMVM) distributions (McNeil et al., 2005) for component errors, referred to as NMVM-MRM. Analogous to the SMSN distributions, the NMVM distributions can accommodate both strong degree of skewness and heavy tails in the data. This class of distributions includes a wide variety of notable members, such as the generalized hyperbolic (GH) distribution (Barndorff-Nielsen and Halgreen, 1977), the normal mean-variance mixture of Birnbaum-Saunders (NMVBS) distribution (Pourmousa et al., 2015), and the normal mean-variance mixture of Lindley (NMVL) distribution (Naderi et al., 2018) as special cases. Importantly, it has a convenient stochastic representation which enables easy implementation of the expectation conditional maximization (ECM) algorithm (Meng and Rubin, 1993) to carry out ML estimation. We develop an efficient expectation conditional maximization either (ECME) algorithm (Liu and Rubin, 1994) for ML estimation of the NMVM-MRM and further offer a simple way of computing the standard errors via the outer product of the scores (Basford et al., 1997).

The paper is structured as follows. Section 2 briefly reviews the NMVM distribution and outlines some of its related properties. Section 3 formulates the proposed model and presents an EM-based procedure for undertaking ML estimation as well as the stopping rule. Section 4 investigates practical issues related to the estimation of standard errors and the specification of valid initial values. Applications to two real data examples are provided in Section 5 for illustration. Four simulation studies are conducted in Section 6 to examine the effectiveness of the proposed methodology. Finally, some concluding remarks are given in Section 7.

2. Normal mean-variance mixture distributions

The NMVM distribution is one of flexible families constructed by scaling the mean and variance of a normal random variable with a positive scale-valued mixing random variable. Specifically, a random variable X is said to follow the NMVM distribution, denoted by $X \sim \text{NMVM}(\mu, \lambda, \sigma^2, \nu)$ henceforth, if it admits the following hierarchical mixing representation:

$$X | (W = w) \sim N(\mu + w\lambda, w\sigma^2) \quad \text{and} \quad W \sim h(w; \nu), \quad (3)$$

where $\mu \in \mathbb{R}$, $\lambda \in \mathbb{R}$, and W is an arbitrary univariate positive random variable with pdf $h(\cdot; \nu)$ indexed by a possibly parameter vector ν . Conditioning on $W = w$, it is readily seen that the mean and variance of X are not fixed. This desirable property helps researchers to derive suitable distributions for modeling data with asymmetry and heavier tails than the normal distribution.

Some special distributions inherited in the NMVM family can be formed by varying the specification of $h(\cdot; \mathbf{v})$. One attractive subclass of the NMVM family, mainly considered in this paper, can be achieved by assuming the generalized inverse Gaussian (GIG; Good (1953)) distribution on W . From (3), X is said to have the GH distribution if W is assumed to have a GIG distribution. More details on the GH formulation are sketched in Appendix A. In Appendix B, we also give an overview on some particular cases of the NMVM class of distributions that are related to the GH distribution. These particular members include the generalized hyperbolic skew- t (GHST), normal inverse Gaussian (NIG), variance-gamma (VG), skew Laplace (SL), NMVBS and NMVL distributions.

3. Model and estimation

3.1. Model specification

According to model (1) coupled with an assumption of $\epsilon_{ij} \sim \text{NMVM}(0, \lambda_i, \sigma_i^2, \mathbf{v}_i)$, the NMVM-MRM is defined henceforth. Let $\Theta = \{\pi, \theta_1, \dots, \theta_g\}$ represent the entire parameters for the NMVM-MRM, where $\pi = (\pi_1, \dots, \pi_{g-1})$ and $\theta_i = \{\beta_i, \lambda_i, \sigma_i^2, \mathbf{v}_i\}$ in which $\mathbf{v}_i = (\kappa_i, \psi_i)^\top$ for VG, $\mathbf{v}_i = \mathbf{v}_i$ for GHST, NIG, NMVBS and NMVL, and lastly $\mathbf{v}_i$ is null for SL. The log-likelihood of parameters Θ for the observed data $\mathbf{y} = (y_1, \dots, y_n)^\top$ can be written as

$$\ell(\Theta | \mathbf{y}) = \sum_{j=1}^n \log \sum_{i=1}^g \pi_i f_{\text{NMVM}}(y_j; \mathbf{x}_j^\top \beta_i, \lambda_i, \sigma_i^2, \mathbf{v}_i), \quad (4)$$

where $f_{\text{NMVM}}(\cdot)$ denotes the pdf of NMVM distribution given in (A.2). As working on mixture modeling, the NMVM-MRM can be formulated by a two-level hierarchical formulation:

$$\begin{aligned} Y_j | (Z_{ij} = 1) &\sim \text{NMVM}(\mathbf{x}_j^\top \beta_i, \lambda_i, \sigma_i^2, \mathbf{v}_i), \\ Z_j &\sim \mathcal{M}(1; \pi_1, \dots, \pi_g). \end{aligned} \quad (5)$$

In light of (3), the NMVM-MRM can be alternatively represented as

$$\begin{aligned} Y_j | (\mathbf{x}_j, W_j = w_j, Z_{ij} = 1) &\sim N(\mathbf{x}_j^\top \beta_i + w_j \lambda_i, w_j \sigma_i^2), \\ W_j | (Z_{ij} = 1) &\sim h(w_j; \mathbf{v}_i), \\ Z_j &\sim \mathcal{M}(1; \pi_1, \dots, \pi_g). \end{aligned} \quad (6)$$

3.2. Identifiability issue

Before developing an EM-type algorithm for the ML parameter estimation of (4), it is important to examine the model identifiability. In the finite mixture framework, a model is said to be identifiable if distinct mixing distributions with finite support correspond to distinct mixtures. Theoretically, a family of distributions $f(x; \theta)$, for $x \in \mathbb{R}$ and $\theta \in \mathbb{R}^d$ is identifiable if two sets of parameters cannot yield the same mixture distribution, that is, $f(x; \theta_1) = f(x; \theta_2)$ for all $x \in \mathbb{R}$ implying that $\theta_1 = \theta_2$. Hennig (2000) discussed the conditions for the identifiability of N-MRMs. Inspired by Hennig (2000), we provide a sufficient condition for the identifiability of the NMVM-MRM (4) in Theorem 1. In this regard, let $\|\cdot\|$ be the Euclidean norm, $\mathbf{x}^-$ denote the covariates excluding the constant for intercept and denote $H = \{\mathbf{x}^- \in \mathbb{R}^p : \mathbf{x}^{-\top} \beta' = a, \text{ with } \mathbb{R}^p \ni \beta' \neq 0\}$ be a $(p-1)$ -dimensional hyperplane, $\mathcal{H}_{p-1}$ the space of $(p-1)$ -dimensional hyperplanes of $\mathbb{R}^p$. Define J to be the minimum number of $(p-1)$ -dimensional hyperplanes, namely

$$J = \min \left\{ k : \{\mathbf{x}_j^- : j \in \mathcal{I}\} \subset \cup_{l=1}^k H_l, \text{ where } H_l \in \mathcal{H}_{p-1} \right\},$$

where the index set $\mathcal{I}$ encompasses the values of distinct covariates.

Theorem 1. Suppose that two pdfs of the NMVM-MRM are parameterized in a different way as

$$f(y; \Theta) = \sum_{i=1}^g \pi_i f_{\text{NMVM}}(y; \mathbf{x}^\top \beta_i, \lambda_i, \sigma_i^2, \mathbf{v}_i) \quad \text{and} \quad f(y; \Theta') = \sum_{i=1}^{g'} \pi'_i f_{\text{NMVM}}(y; \mathbf{x}^\top \beta'_i, \lambda'_i, \sigma'^2_i, \mathbf{v}'_i).$$

For the considered examples of the NMVM distribution in Section 2, if conditions $g < J$ and $\|\beta_r - \beta_l\|^2 \neq 0$ for $r \neq l \in \{1, \dots, g\}$ hold, then the equality $f(y; \Theta) = f(y; \Theta')$ implies that $g = g'$, and there are $r, l \in \{1, \dots, g\}$ such that $\beta_r = \beta'_l$, $\sigma_r^2 = \sigma'^2_l$, $\lambda_r = \lambda'_l$, and $\mathbf{v}_r = \mathbf{v}'_l$.

Proof. We first note that the finite mixture of our considered special cases of the NMVM distributions is identifiable, see, for instance, Naderi et al. (2019) for the identifiability issue of the finite mixture of the NMVBS distributions. Now, if conditions $g < J$ and $\|\beta_r - \beta_l\|^2 \neq 0$ are fulfilled, the proposed model is identifiable conditionally to each covariate point $\mathbf{x}$, and the proof can be obtained directly from the proof of Theorem 2.2 of Hennig (2000). $\square$

3.3. ML parameter estimation

To compute the ML estimates of parameters of the NMVM-MRM, we adopt the ECME algorithm as proposed by Liu and Rubin (1994), which is an efficient variant of the EM algorithm (Dempster et al., 1977) and a simple extension of the ECM algorithm (Meng and Rubin, 1993) with the maximization (M) step of EM replaced by a sequence of computationally simpler conditional maximization (CM) steps that maximize either the Q -function (CMQ-step) or the corresponding constrained actual likelihood function (CML-step). From (6), the complete-data log-likelihood function of Θ associated with the observed data $\mathbf{y}$, the scaling variables $\mathbf{W} = (W_1, \dots, W_n)^\top$ and allocation indicators $\mathbf{Z} = \{Z_1, \dots, Z_n\}$, omitting terms irrelevant to Θ , is

$$\ell_c(\Theta | \mathbf{y}, \mathbf{W}, \mathbf{Z}) = \sum_{j=1}^n \ell_{cj}(\Theta | \mathbf{y}_{cj}), \quad (7)$$

where $\mathbf{y}_{cj} = (y_j, W_j, \mathbf{Z}_j^\top)^\top$ denotes the complete-data set for the j th individual, and

$$\begin{aligned} \ell_{cj}(\Theta | \mathbf{y}_{cj}) = & \sum_{i=1}^g Z_{ij} \left[\log \pi_i - \frac{1}{2} \log \sigma_i^2 + \log h(W_j; \mathbf{v}_i) \right. \\ & \left. - \frac{1}{2\sigma_i^2} \left\{ W_j^{-1} (y_j - \mathbf{x}_j^\top \beta_i)^2 - 2\lambda_i (y_j - \mathbf{x}_j^\top \beta_i) + W_j \lambda_i^2 \right\} \right] \end{aligned} \quad (8)$$

is the individual complete-data log-likelihood corresponding to $\mathbf{y}_{cj}$.

The ECME algorithm for ML estimation of the NMVM-MRM proceeds as follows.

E-step: At the k th iteration, we compute the Q -function, defined as the expected value of complete-data log-likelihood (7) with Θ evaluated at $\hat{\Theta}^{(k)}$. This yields

$$\begin{aligned} Q(\Theta | \hat{\Theta}^{(k)}) = & \sum_{i=1}^g \sum_{j=1}^n \hat{z}_{ij}^{(k)} \left[\log \pi_i - \frac{1}{2} \log \sigma_i^2 + \hat{\gamma}_{ij}^{(k)}(\mathbf{v}_i) \right. \\ & \left. - \frac{1}{2\sigma_i^2} \left\{ \hat{t}_{ij}^{(k)} (y_j - \mathbf{x}_j^\top \beta_i)^2 - 2\lambda_i (y_j - \mathbf{x}_j^\top \beta_i) + \hat{w}_{ij}^{(k)} \lambda_i^2 \right\} \right], \end{aligned} \quad (9)$$

where $\hat{\gamma}_{ij}^{(k)}(\mathbf{v}_i) = E[\log h(W_j; \mathbf{v}_i) | y_j, Z_{ij} = 1, \hat{\Theta}^{(k)}]$, $\hat{w}_{ij}^{(k)} = E[W_j | y_j, Z_{ij} = 1, \hat{\Theta}^{(k)}]$, $\hat{t}_{ij}^{(k)} = E[W_j^{-1} | y_j, Z_{ij} = 1, \hat{\Theta}^{(k)}]$, and

$$\hat{z}_{ij}^{(k)} = E[Z_{ij} | y_j, \hat{\Theta}^{(k)}] = \frac{\hat{\pi}_i^{(k)} f_{\text{NMVM}}(y_j; \mathbf{x}_j^\top \hat{\beta}_i^{(k)}, \hat{\lambda}_i^{(k)}, \hat{\sigma}_i^{2(k)}, \hat{\mathbf{v}}_i^{(k)})}{\sum_{h=1}^g \hat{\pi}_h^{(k)} f_{\text{NMVM}}(y_j; \mathbf{x}_j^\top \hat{\beta}_h^{(k)}, \hat{\lambda}_h^{(k)}, \hat{\sigma}_h^{2(k)}, \hat{\mathbf{v}}_h^{(k)})} \quad (10)$$

represents the posterior probability that y_j is assigned to the i th mixture component. Apart from $\hat{z}_{ij}^{(k)}$, the exact expressions for these conditional expectations rely on the choice of distributions for W_j . Notice that $\hat{w}_{ij}^{(k)}$ and $\hat{t}_{ij}^{(k)}$ can be easily evaluated using the results stated in Appendix B. For updating $\hat{\mathbf{v}}_i^{(k)}$, the calculation of $\hat{\gamma}_{ij}^{(k)}(\mathbf{v}_i)$ could be very complicated, but it can be avoided by adopting the CML-step of the ECME algorithm.

CMQ-step: Maximizing (9) over parameters leads to the following CM estimators:

$$\begin{aligned} \hat{\pi}_i^{(k+1)} &= \frac{\hat{n}_i^{(k)}}{n}, \quad \hat{\beta}_i^{(k+1)} = \left[\sum_{j=1}^n \hat{z}_{ij}^{(k)} \hat{t}_{ij}^{(k)} \mathbf{x}_j \mathbf{x}_j^\top \right]^{-1} \left[\sum_{j=1}^n \hat{z}_{ij}^{(k)} (\hat{t}_{ij}^{(k)} y_j - \hat{\lambda}_i^{(k)}) \mathbf{x}_j \right], \\ \hat{\lambda}_i^{(k+1)} &= \frac{\sum_{j=1}^n \hat{z}_{ij}^{(k)} (y_j - \mathbf{x}_j^\top \hat{\beta}_i^{(k+1)})}{\sum_{j=1}^n \hat{z}_{ij}^{(k)} \hat{w}_{ij}^{(k)}}, \\ \hat{\sigma}_i^{2(k+1)} &= \frac{1}{\hat{n}_i^{(k)}} \sum_{j=1}^n \hat{z}_{ij}^{(k)} \left\{ \hat{t}_{ij}^{(k)} (y_j - \mathbf{x}_j^\top \hat{\beta}_i^{(k+1)})^2 - 2\hat{\lambda}_i^{(k+1)} (y_j - \mathbf{x}_j^\top \hat{\beta}_i^{(k+1)}) + \hat{w}_{ij}^{(k)} \hat{\lambda}_i^{2(k+1)} \right\}, \end{aligned}$$

and

Algorithm 1 Implementation procedure of the ECME algorithm for fitting the NMVM-MRM.

```

1: procedure NMVM-MRM( $y, \mathbf{x}$ )
2:   inputs:  $\{y_i, \mathbf{x}_i\}_{i=1}^n$  - the set of input data;  $g$  - the number of components;
3:    $K_{\max} = 2000$  - the maximum number of iteration;  $\varepsilon = 10^{-5}$  - the prespecified tolerance.
4:   initialize: Obtain the starting value  $\hat{\Theta}^{(0)}$ .
5:   Compute the initial log-likelihood as  $\ell_{\text{obs}}^{(0)} = \ell(\hat{\Theta}^{(0)} | y)$ .
6:   Set "Convergence = False".
7:   for  $k = 0$  to  $K_{\max}$  do
8:     for  $j = 1$  to  $n$  do
9:       for  $i = 1$  to  $g$  do
10:        Obtain latent and missing information  $\hat{z}_{ij}^{(k)}, \hat{w}_{ij}^{(k)}, \hat{t}_{ij}^{(k)}$ , and  $\hat{\gamma}_{ij}^{(k)}(\mathbf{v}_i)$ .
11:      end for
12:    end for
13:    for  $i = 1$  to  $g$  do
14:      Update the parameters  $\hat{\pi}_i^{(k+1)}, \hat{\lambda}_i^{(k+1)}, \hat{\sigma}_i^{2(k+1)}, \hat{\beta}_i^{(k+1)}$ , and  $\hat{\mathbf{v}}_i^{(k+1)}$ .
15:      Set  $\hat{\Theta}^{(k+1)} = \{\hat{\pi}_i^{(k+1)}, \hat{\theta}_1^{(k+1)}, \dots, \hat{\theta}_g^{(k+1)}\}$ .
16:    end for
17:    Compute the updated log-likelihood  $\ell_{\text{obs}}^{(k+1)} = \ell(\hat{\Theta}^{(k+1)} | y)$ .
18:    Compute the Aitken's acceleration factor and set "Convergence = True" if  $\ell_{\infty}^{(k+1)} - \ell_{\text{obs}}^{(k)} < \varepsilon$ .
19:    if Convergence = True then
20:      break
21:    end if
22:  end for
23:  return  $\ell_{\text{obs}}^{(k+1)}$  and  $\hat{\Theta}^{(k+1)}$ .
24: end procedure

```

$$\hat{\mathbf{v}}_i^{(k+1)} = \begin{cases} (\overline{z\mathbf{w}}_i^{(k)})^{-1} & \text{for NIG-MRM,} \\ \sqrt{\overline{z\mathbf{w}}_i^{(k)} + z\mathbf{t}_i^{(k)} - 2} & \text{for NMVBS-MRM,} \\ \frac{\sqrt{8\overline{z\mathbf{w}}_i^{(k)} + (1 - \overline{z\mathbf{w}}_i^{(k)})^2} + (1 - \overline{z\mathbf{w}}_i^{(k)})}{2\overline{z\mathbf{w}}_i^{(k)}} & \text{for NMVL-MRM,} \end{cases}$$

where $\hat{n}_i^{(k)} = \sum_{j=1}^n \hat{z}_{ij}^{(k)}$, $\overline{z\mathbf{w}}_i^{(k)} = \sum_{j=1}^n \hat{z}_{ij}^{(k)} \hat{w}_{ij}^{(k)} / \hat{n}_i^{(k)}$ and $\overline{z\mathbf{t}}_i^{(k)} = \sum_{j=1}^n \hat{z}_{ij}^{(k)} \hat{t}_{ij}^{(k)} / \hat{n}_i^{(k)}$.

CML-step: Because the calculation of $\hat{\gamma}_{ij}^{(k)}(\mathbf{v}_i)$ for the VG-MRM and GHST-MRM is very complicated, we update $\hat{\mathbf{v}}_i^{(k)}$, for ease of implementation, by maximizing the constrained actual observed log-likelihood function corresponding to (5), namely

$$\hat{\mathbf{v}}_i^{(k+1)} = \arg \max_{\mathbf{v}_i} \left\{ \sum_{j=1}^n \hat{z}_{ij}^{(k)} \log f_{\text{NMVM}} \left(y_j; \mathbf{x}_j^\top \hat{\beta}_i^{(k+1)}, \hat{\lambda}_i^{(k+1)}, \hat{\sigma}_i^{2(k+1)}, \mathbf{v}_i \right) \right\}.$$

The E- and CM-steps of the above ECME algorithm are iterated until a suitable stopping criterion is satisfied. To monitor the convergence, we adopt the Aitken acceleration method (Aitken, 1927) to avoid an indication of lack of progress of the algorithm. The asymptotic log-likelihood at iteration $k+1$ is estimated by $\ell_{\infty}^{(k+1)} = \ell_{\text{obs}}^{(k)} + (\ell_{\text{obs}}^{(k+1)} - \ell_{\text{obs}}^{(k)}) / (1 - a^{(k)})$, where $\ell_{\text{obs}}^{(k)}$ stands for the observed log-likelihood and $a^{(k)} = (\ell_{\text{obs}}^{(k+1)} - \ell_{\text{obs}}^{(k)}) / (\ell_{\text{obs}}^{(k)} - \ell_{\text{obs}}^{(k-1)})$ is the Aitken's acceleration factor calculated as the ratio of successive increments evaluated at iteration k . In our study, the algorithm is declared to reach convergence when either the criterion $\ell_{\infty}^{(k+1)} - \ell_{\text{obs}}^{(k)} < 10^{-5}$ or the maximum number of iterations $K_{\max} = 2000$ has been reached. For a detailed guide of implementation, a pseudocode of the above ECME algorithm is summarized in Algorithm 1.

4. Computational aspects

4.1. Estimation of standard errors

We describe, herein, how to compute the standard errors of ML parameter estimates for the NMVM-MRM. Under regularity conditions, the asymptotic covariance matrix of $\hat{\Theta}$ can be approximated by the inverse of the empirical information matrix, defined as

$$\mathbf{I}_e(\hat{\Theta} | y) = \sum_{j=1}^n \mathbf{s}(y_j | \hat{\Theta}) \mathbf{s}^\top(y_j | \hat{\Theta}), \quad (11)$$

where $\mathbf{s}(y_j | \Theta) = \partial \log f(y_j | \Theta) / \partial \Theta$ is the individual score, but its explicit form is not obtainable by directly taking the partial derivative for the observed log-likelihood function. Following Louis (1982), the individual score can be alternatively computed by taking the expectation for the partial derivative of the individual complete-data log-likelihood function given in (8), i.e.,

$$\mathbf{s}(y_j | \hat{\Theta}) = E \left[\frac{\partial \ell_{cj}(\Theta | \mathbf{y}_{cj})}{\partial \Theta} \middle| y_j, \hat{\Theta} \right]. \quad (12)$$

Using the fundamental differentiation for (12),

$$\mathbf{s}(y_j | \hat{\Theta}) = (\hat{s}_{j,\pi_1}, \dots, \hat{s}_{j,\pi_{g-1}}, \hat{s}_{j,\beta_1}^\top, \dots, \hat{s}_{j,\beta_g}^\top, \hat{s}_{j,\lambda_1}, \dots, \hat{s}_{j,\lambda_g}, \hat{s}_{j,\sigma_1^2}, \dots, \hat{s}_{j,\sigma_g^2}, \hat{s}_{j,\mathbf{v}_1}^\top, \dots, \hat{s}_{j,\mathbf{v}_g}^\top)^\top$$

is a $m \times 1$ vector comprising the following explicit expressions:

$$\hat{s}_{j,\pi_i} = E \left[\frac{\partial \ell(\Theta | \mathbf{y}_{cj})}{\partial \pi_i} \middle| y_j, \hat{\Theta} \right] = \frac{\hat{z}_{ij}}{\pi_i} - \frac{\hat{z}_{gj}}{\pi_g},$$

for $i = 1, \dots, g-1$, and

$$\hat{s}_{j,\beta_i}^\top = E \left[\frac{\partial \ell(\Theta | \mathbf{y}_{cj})}{\partial \beta_i} \middle| y_j, \hat{\Theta} \right] = \frac{\hat{z}_{ij}}{\hat{\sigma}_i^2} (\hat{t}_{ij}(y_j - \mathbf{x}_j^\top \hat{\beta}_i) - \hat{\lambda}_i) \mathbf{x}_j^\top,$$

$$\hat{s}_{j,\lambda_i} = E \left[\frac{\partial \ell(\Theta | \mathbf{y}_{cj})}{\partial \lambda_i} \middle| y_j, \hat{\Theta} \right] = \frac{\hat{z}_{ij}}{\hat{\sigma}_i^2} (y_j - \mathbf{x}_j^\top \hat{\beta}_i - \hat{w}_{ij} \hat{\lambda}_i),$$

$$\hat{s}_{j,\sigma_i^2} = E \left[\frac{\partial \ell(\Theta | \mathbf{y}_{cj})}{\partial \sigma_i^2} \middle| y_j, \hat{\Theta} \right] = \frac{\hat{z}_{ij}}{2\hat{\sigma}_i^2} \left(\frac{1}{\hat{\sigma}_i^2} \left\{ \hat{t}_{ij} (y_j - \mathbf{x}_j^\top \hat{\beta}_i)^2 - 2\hat{\lambda}_i (y_j - \mathbf{x}_j^\top \hat{\beta}_i) + \hat{w}_{ij} \hat{\lambda}_i^2 \right\} - 1 \right),$$

for $i = 1, \dots, g$, where $\hat{z}_{ij}$, $\hat{w}_{ij}$ and $\hat{t}_{ij}$ are defined the same as those in (9) and (10) when the algorithm reaches convergence.

As for $\hat{s}_{j,\mathbf{v}_i}$, the expressions depend on the distribution of W_j . For some particular cases considered in this paper, except for SL which is null, we have

$$\hat{s}_{j,\mathbf{v}_i}^\top = \begin{cases} \hat{z}_{ij} \hat{\Upsilon}'_{ij}(\hat{\mathbf{v}}_i) & \text{for GHST and VG,} \\ \hat{z}_{ij} (1 - \hat{\mathbf{v}}_i \hat{w}_{ij}) & \text{for NIG,} \\ \hat{z}_{ij} (\hat{\mathbf{v}}_i^{-3} (\hat{w}_{ij} + \hat{t}_{ij} - 2) - \hat{\mathbf{v}}_i^{-1}) & \text{for NMVBS,} \\ \hat{z}_{ij} \left\{ \frac{2 + \hat{\mathbf{v}}_i}{\hat{\mathbf{v}}_i(1 + \hat{\mathbf{v}}_i)} - \hat{w}_{ij} \right\} & \text{for NMVL,} \end{cases}$$

where $\hat{\Upsilon}'_{ij}(\hat{\mathbf{v}}_i) = E[\partial \log h(W_j; \mathbf{v}_i) / \partial \mathbf{v}_i | y_j, Z_{ij} = 1, \hat{\Theta}]$. As a result, the standard errors for a specific parameter estimate $\hat{\theta}_r$ can be approximated by

$$se(\hat{\theta}_r) \approx \sqrt{[I_e^{-1}(\hat{\Theta} | \mathbf{y})]_{rr}}, \quad (13)$$

where $[I_e^{-1}(\hat{\Theta} | \mathbf{y})]_{rr}$ is the (r, r) th entry of the inverse of (11).

4.2. Initialization

Since the log-likelihood function of the MRM typically exhibits multiple local maxima, the iterative procedure is likely to get stuck at local maxima or saddle points, especially when the initial values depart far from the real parameter values. Meanwhile, good initial values may accelerate the convergence and increase the probability of finding the global solution. Thus, it is greatly important to try a variety of sensible starting values for the proposed ECME algorithm. Herein, we recommend a simple way of obtaining valid initial values as follows:

Step 1. Partition the data $(y_1, \mathbf{x}_1), \dots, (y_n, \mathbf{x}_n)$ into g groups by performing the hard-assignment clustering method, such as the k -means clustering algorithm (Hartigan and Wong, 1979) or its robust variants, e.g., the *trimmed k-means* (García-Escudero and Gordaliza, 1999) and *MinMax k-means clustering* (Tzortzis and Likas, 2014) algorithms.

Step 2. Given each partition of the data, the initial value for $\hat{\pi}_i^{(0)}$ is determined to be the proportion of data points belonging to the i th group. We then proceed the least trimmed squares (Rousseeuw and Leroy, 2005) to obtain $\hat{\beta}_i^{(0)}$ and use the mean squared errors of residuals to initialize $\hat{\sigma}_i^{2(0)}$.

Table 1

Model comparisons for fitting various MRMs to the tone perception data. The lowest score of each criterion is indicated in bold.

Criteria	MRM						
	N	SL	VG	NIG	GHST	NMVBS	NMVL
m	7	9	13	11	11	11	11
$\ell_{\max}$	141.198	174.290	217.991	226.127	231.813	232.974	173.107
AIC	−268.397	−330.581	−409.981	−430.254	−441.625	−443.948	−324.214
BIC	−247.322	−303.485	−370.843	−397.137	−408.508	−410.831	−291.097
CAIC	−267.608	−329.295	−404.138	−428.341	−439.712	−442.035	−322.301
ABIC	−269.476	−331.968	−407.305	−431.950	−443.321	−445.644	−325.910

Step 3. The initial values for the skewness $\hat{\lambda}_i^{(0)}$ and shape $\hat{\nu}_i^{(0)}$ parameters are obtained by fitting the corresponding NMVM distribution to residuals in the preceding step.

Step 4. Go to Step 1 to create a new partition of data into g groups and execute Step 2 to extract a different set of initial values.

5. Applications to real data

In this section, we perform the analyses of two real data examples to illustrate the usefulness of our proposed NMVM-MRMs. For comparison purposes, the traditional N-MRM approach is also fitted to these datasets. To assess the appropriateness of the fitted models, we adopt four penalized-likelihood information criteria, including the Akaike information criterion (Akaike, 1998, AIC), Bayesian information criterion (Schwarz, 1978, BIC), corrected AIC (Bedrick and Tsai, 1994, CAIC) and adjusted BIC (Scllove, 1987, ABIC), defined as $c(n, m) - 2\ell_{\max}$, where m is the number of free parameters in the candidate model, and $\ell_{\max}$ is the maximized log-likelihood value. The penalty term $c(n, m)$ is $2m$ for AIC, $m \log(n)$ for BIC, $2nm/(n - m - 1)$ for CAIC, and $m \log((n + 2)/24)$ for ABIC, respectively. The model with smaller values of these measures is considered to provide a better fit to the data.

To examine the influence of outlying observations on the estimation of model parameters, we calculate the likelihood distance (LD) measure (Cook and Weisberg, 1982), defined as

$$LD_s = 2|\ell(\hat{\Theta} | \mathbf{y}) - \ell(\hat{\Theta}_{-s} | \mathbf{y})|, \quad (14)$$

where $\hat{\Theta}_{-s}$ is the ML estimates of Θ excluding the s th observation in the data. The s th individual is determined as an outlier if the value of LD_s is relatively large.

5.1. Example 1: analysis of the tone perception data

Our first illustration concerns the tone perception data (Cohen, 1984) containing 150 trials from the same musician. The data were collected from an experiment in which a pure fundamental tone determined by a ‘stretching ratio’ (actual tone) was first provided to a trained musician. Afterwards, the musician was asked to tune an adjustable tone to one octave above the fundamental tone and the ‘tuning ratio’ (perceived tone), measured as the adjusted tone divided by the fundamental tone, was recorded. The goal of this experiment was to see how this actual stretching ratio (x) affects the perception of the tone (y) and to determine the reasonability of the two musical perception theories. The data were previously analyzed by Viele and Tong (2002) and Hunter and Young (2012) via the semiparametric and Bayesian N-MRM approaches, respectively.

Fig. 1 displays the scatter plot, histograms and boxplots for the actual versus perceived tone ratios of the data. It can be observed that there are two apparent linear patterns between two variables. Moreover, the marginal histograms and boxplots indicate the existence of strong non-normal features in the data. Table 1 presents the ML fitting results obtained by a 2-component N-MRM and six candidate models in the NMVM-MRM class, including SL, VG, NIG, GHST, NMVBS and NMVL, to the data. The results reveal that the six NMVM-MRMs are all superior to the N-MRM, which has the worst fit due to a lack of ability against non-normal features. The NMVBS-MRM provides the best fit among all candidates, no matter which criterion is used. As an illustration, the regression lines estimated by the NMVBS-MRM and N-MRM are also superimposed in Fig. 1. It appears that the two fitted models provide somewhat different regression lines.

Fig. 2 depicts the LD measures calculated by using (14) under four chosen MRMs (N, NIG, GHST and NMVBS) from which the latter three NMVM-MRMs provide a relatively better fit in terms of BIC. It is interesting to mention that all selected models identify case #60 as an extreme outlier, which is marked by a red triangle on the top right corner of Fig. 1. Furthermore, we observe that cases #56, #85, and #147 under N-MRM and NIG-MRM could be considered as mild outliers, highlighted in Fig. 1 by yellow squares falling between the two regression lines with x ranging from 2.5 to 3. On the contrary, these three mild outliers show relatively little effect on the parameter estimation for GHST-MRM and NMVBS-MRM, showing an empirical evidence that the two best fitted models can offer more robustness against highly influential observations in this data example.

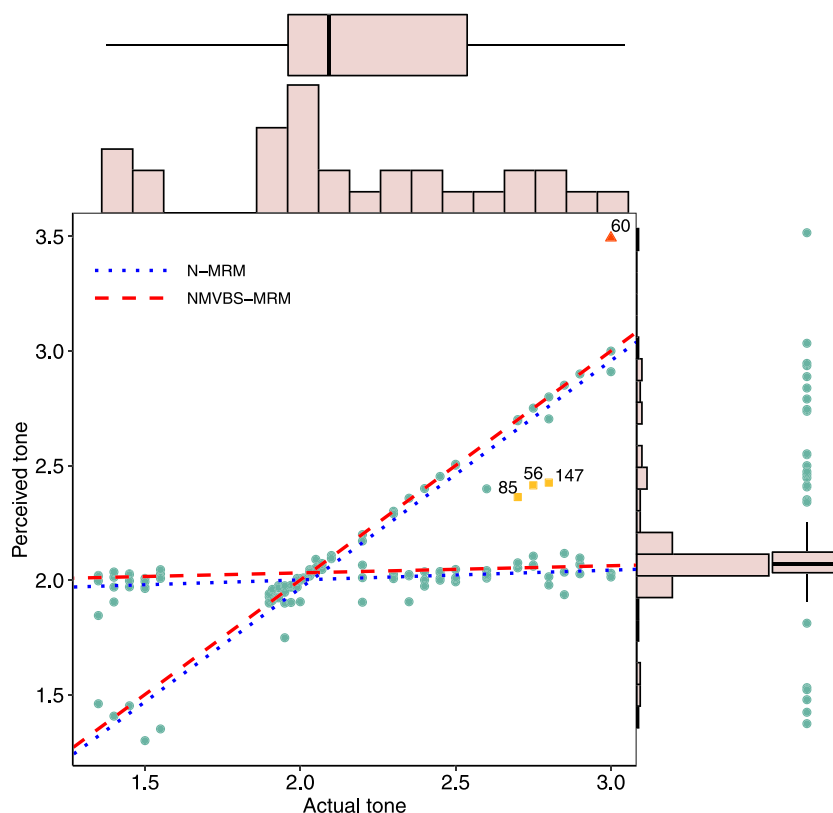


Fig. 1. Tone perception data. The scatter plot overlaid with regression lines obtained from the fitted NMVBS-MRM and N-MRM together with their marginal histograms and boxplots. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Table 2

Model comparisons for fitting various MRMs to the *Pinus nigra* data. The lowest score of each criterion is indicated in bold. NA (not available) is meant for non-convergence.

Criteria	MRM						
	N	SL	VG	NIG	GHST	NMVBS	NMVL
m	11	14	20	17	17	17	17
$\ell_{\max}$	−854.694	−822.085	NA	−812.630	−813.599	−816.652	−821.924
AIC	1731.388	1672.171	NA	1659.261	1661.197	1667.304	1677.846
BIC	1774.105	1726.537	NA	1725.277	1727.214	1733.320	1743.862
CAIC	1732.149	1673.392	NA	1661.056	1662.992	1669.099	1679.640
ABIC	1739.207	1682.122	NA	1671.345	1673.281	1679.388	1689.930

5.2. Example 2: analysis of the *Pinus nigra* data

As a second example, we consider the *Pinus nigra* data which consist of diameters in the tree foot (in millimeters) and heights (in meters; trunk height without branches) of $n = 359$ trees located in the north of Palencia, a province of Spain. In the forestry and forest industry, a classical problem is to search for reliable models that allow for predicting the height of a tree from its diameter (Schreuder and Hafley, 1977). Moreover, García-Escudero et al. (2010) and Mazza and Punzo (2017) have conducted different robust approaches to modeling the relation between the response variable “height” (y) and the covariate “diameter” (x).

Observing the scatter plot in Fig. 3, there exist three distinct linear groups apart from one isolated tree on the bottom right corner and a small group of trees on the top right corner. This can also be verified from some earlier studies (García-Escudero et al., 2010; Mazza and Punzo, 2017). With g a priori fixed at 3, we consider the fitting of NMVM-MRMs and N-MRM to this dataset. By comparing the results reported in Table 2, the NIG-MRM attains the best fit. The VG-MRM, which owns the most parameters, suffers from a certain non-convergence problem. Analogous to the first example, we calculate the LD measures of four selected MRMs using (14). Looking at Fig. 4, all models strongly indicated that the first nine trees are suspect outliers, marked as red triangles in Fig. 3, where the regression lines obtained from the NIG-MRM and N-MRM are also superimposed. It is encouraging to see that the regression lines of the fitted NIG-MRM can adapt the trend of grouping patterns more suitably than those of N-MRM.

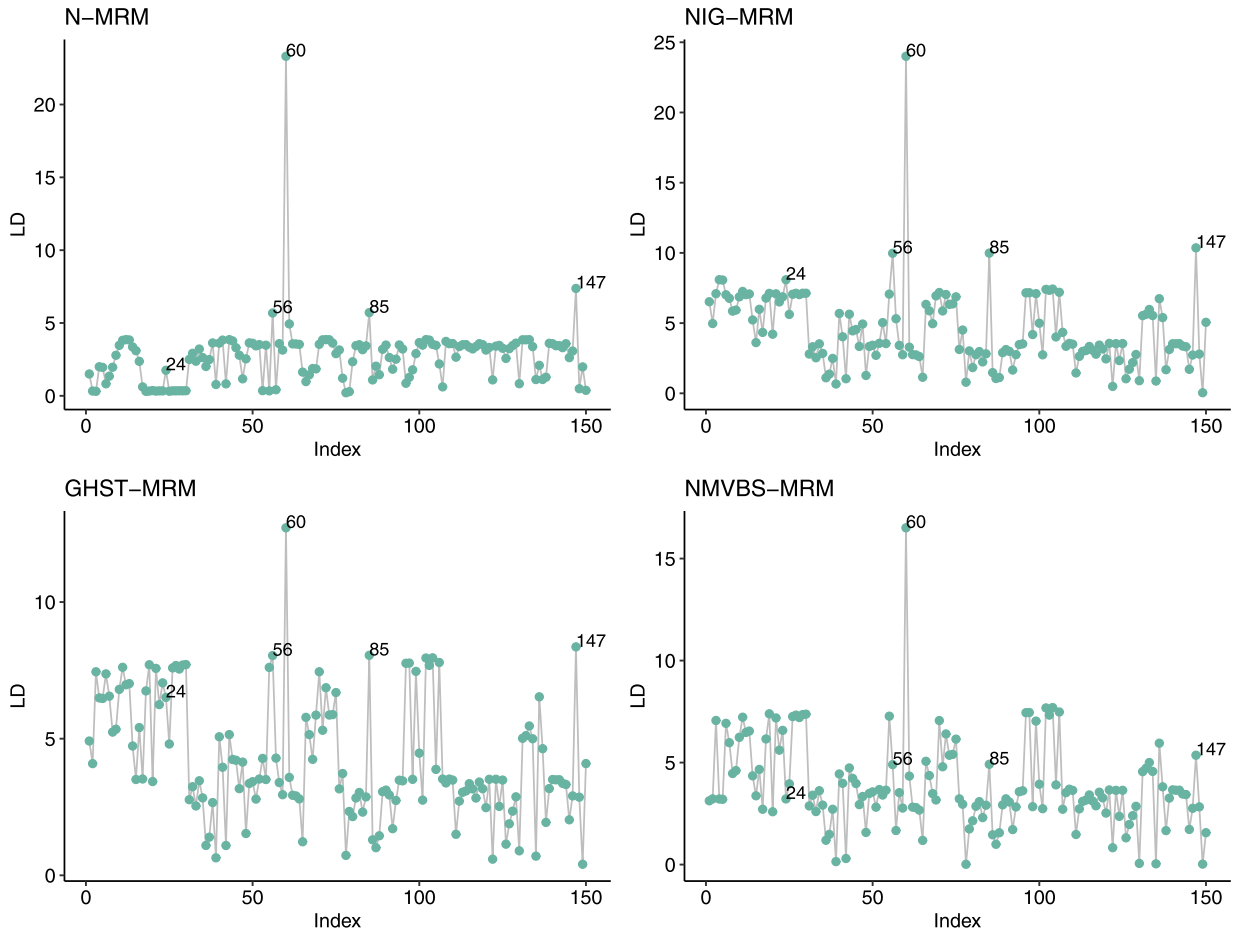


Fig. 2. Plot of LD measures for the tone perception data fitted by four selected MRMs.

6. Simulation experiments

6.1. Finite sample properties

The first simulation study is undertaken to examine the performance of the ML estimates obtained through the ECME algorithm and their standard errors via the information-based method described in Section 4.1. As an illustration, 500 Monte Carlo (MC) samples of size $n = 100, 500, 1000$ and 2000 are generated from the NMVBS-MRM with $g = 2$. The presumed parameters are $\lambda_1 = \nu_1 = 2$, $\lambda_2 = \nu_2 = 3$, $\pi_1 = 0.4$, $\beta_1 = (0.5, -1, -2, -3)^\top$, $\beta_2 = (-1, 1, 2, 3)^\top$, $\sigma_1^2 = 1$, and $\sigma_2^2 = 2$. The design matrix is specified as $\mathbf{x}_j = (1, x_{1j}, x_{2j}, x_{3j})^\top$ in which x_{1j} , x_{2j} , and x_{3j} are randomly drawn from $U(1, 5)$, $U(-2, 2)$, and $U(1, 4)$, respectively, where $U(a, b)$ represents the uniform distribution on the interval (a, b) . To measure the estimation accuracy, we compute the absolute relative bias (ARB) and the root mean squared error (RMSE):

$$\text{ARB} = \frac{1}{500} \sum_{r=1}^{500} \left| \frac{\hat{\theta}_r - \theta_{\text{true}}}{\theta_{\text{true}}} \right| \quad \text{and} \quad \text{RMSE} = \sqrt{\frac{1}{500} \sum_{r=1}^{500} (\hat{\theta}_r - \theta_{\text{true}})^2},$$

where $\hat{\theta}_r$ denotes the ML estimate of a specific parameter at the r th replication and θ_{true} is its true value. In addition, we compute the standard deviation (STD) of ML estimates across 500 replications together with the average values of their approximate standard errors (ASE) based on the empirical information matrix. They are calculated as

$$\text{STD} = \sqrt{\frac{1}{499} \sum_{r=1}^{500} \left(\hat{\theta}_r - \frac{1}{500} \sum_{r=1}^{500} \hat{\theta}_r \right)^2} \quad \text{and} \quad \text{ASE} = \frac{1}{500} \sum_{r=1}^{500} \text{se}(\hat{\theta}_r).$$

It can be clearly seen from Fig. 5 that both magnitudes of ARB and RMSE tend to decrease toward zero by increasing n , showing empirically the consistency of the ML estimators. Moreover, numerical results reported in Table 3 indicate that

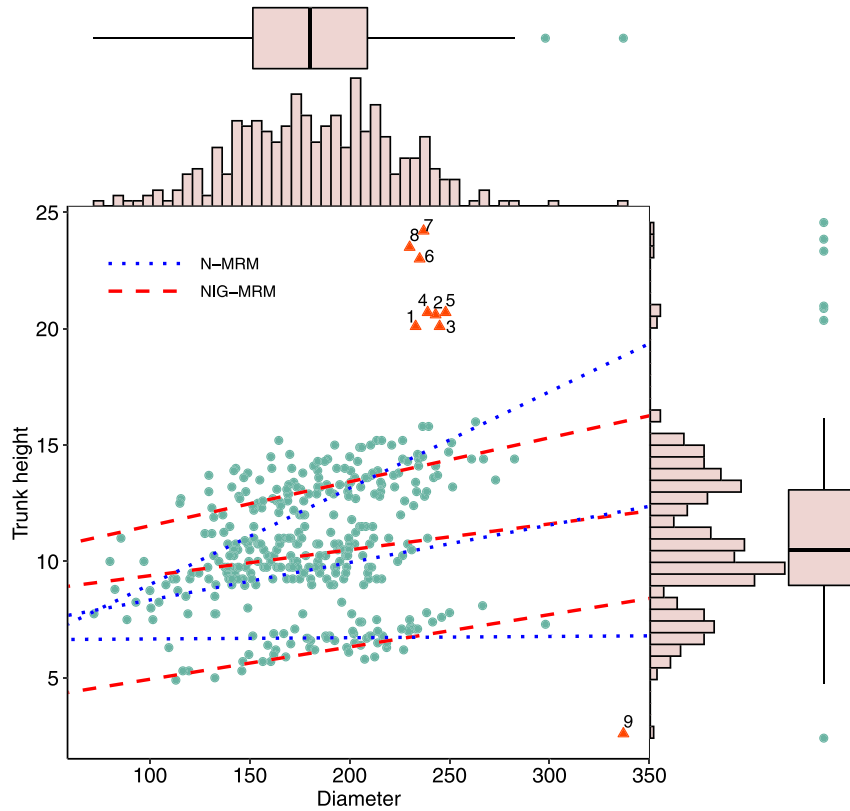


Fig. 3. Pinus nigra data. The scatter plot overlaid with regression lines obtained from the fitted NIG-MRM and N-MRM together with their marginal histograms and boxplots.

Table 3

Simulation results for assessing the asymptotic standard errors and empirical standard deviations of parameters estimates across various sample sizes.

n	Measure	β_{01}	β_{11}	β_{21}	β_{31}	β_{02}	β_{12}	β_{22}	β_{32}	σ_1^2	σ_2^2	λ_1	λ_2	ν_1	ν_2	π_1
100	STD	0.8395	0.2292	0.2288	0.2762	0.7594	0.1639	0.1700	0.2394	1.2185	1.8765	0.5383	0.6077	1.5620	0.9457	0.0602
	ASE	1.3672	0.3564	0.3616	0.1793	0.6423	0.1202	0.1255	0.1445	1.7819	1.3951	0.2734	0.4591	1.0956	1.2274	0.0631
500	STD	0.2864	0.0585	0.0573	0.0710	0.2328	0.0508	0.0505	0.0626	0.4519	0.6453	0.2257	0.2334	0.3809	0.2429	0.0312
	ASE	0.4256	0.0836	0.0912	0.0522	0.1999	0.0343	0.0401	0.0453	0.6215	0.4971	0.1863	0.1813	0.2971	0.3391	0.0331
1000	STD	0.2000	0.0378	0.0376	0.0547	0.1687	0.0320	0.0341	0.0464	0.2876	0.4116	0.1509	0.1584	0.2636	0.1915	0.0217
	ASE	0.2613	0.0399	0.0402	0.0515	0.1373	0.0300	0.0319	0.0404	0.3194	0.3815	0.1406	0.1434	0.2099	0.2483	0.0223
2000	STD	0.1280	0.0276	0.0256	0.0340	0.1098	0.0237	0.0235	0.0302	0.2013	0.3068	0.0953	0.1074	0.1812	0.1159	0.0145
	ASE	0.1301	0.0291	0.0257	0.0331	0.0986	0.0232	0.0228	0.0301	0.2210	0.2911	0.0917	0.0979	0.1794	0.1209	0.0146

the values of STD and ASE are getting closer as n increases, signifying that the validity of (13) for calculating the standard errors of the ML estimates. We mention by passing that simulation studies were also conducted for other NMVM-MRMs. The results are similar and hence are not reported for the sake of brevity.

6.2. Sensitivity of the EM estimates in presence of outliers

The second simulation aims at studying the robustness aspects of the NMVM-MRM on parameter estimation when the simulated data are contaminated with a single outlier. To conduct the experiment, a total of 100 Monte Carlo samples of size $n = 1000$ are generated from the 3-component SN-MRM (Liu and Lin, 2014), defined as model (1) together with $\epsilon_{ij} \sim SN(0, \sigma_i^2, \lambda_i)$, for $i = 1, 2, 3$ and $\pi_i = 1/3$, where the true parameters to have parallel and concurrent configurations are specified as

$$\begin{aligned} \text{parallel: } & \beta_1 = (-3, 1)^\top, \beta_2 = (3, 1)^\top, \beta_3 = (0.5, 1)^\top, \sigma_1^2 = 0.2, \sigma_2^2 = 0.4, \sigma_3^2 = 0.6; \\ \text{concurrent: } & \beta_1 = (-1, 3)^\top, \beta_2 = (3, -1)^\top, \beta_3 = (3, 1)^\top, \sigma_1^2 = 0.6, \sigma_2^2 = 0.9, \sigma_3^2 = 0.3. \end{aligned}$$

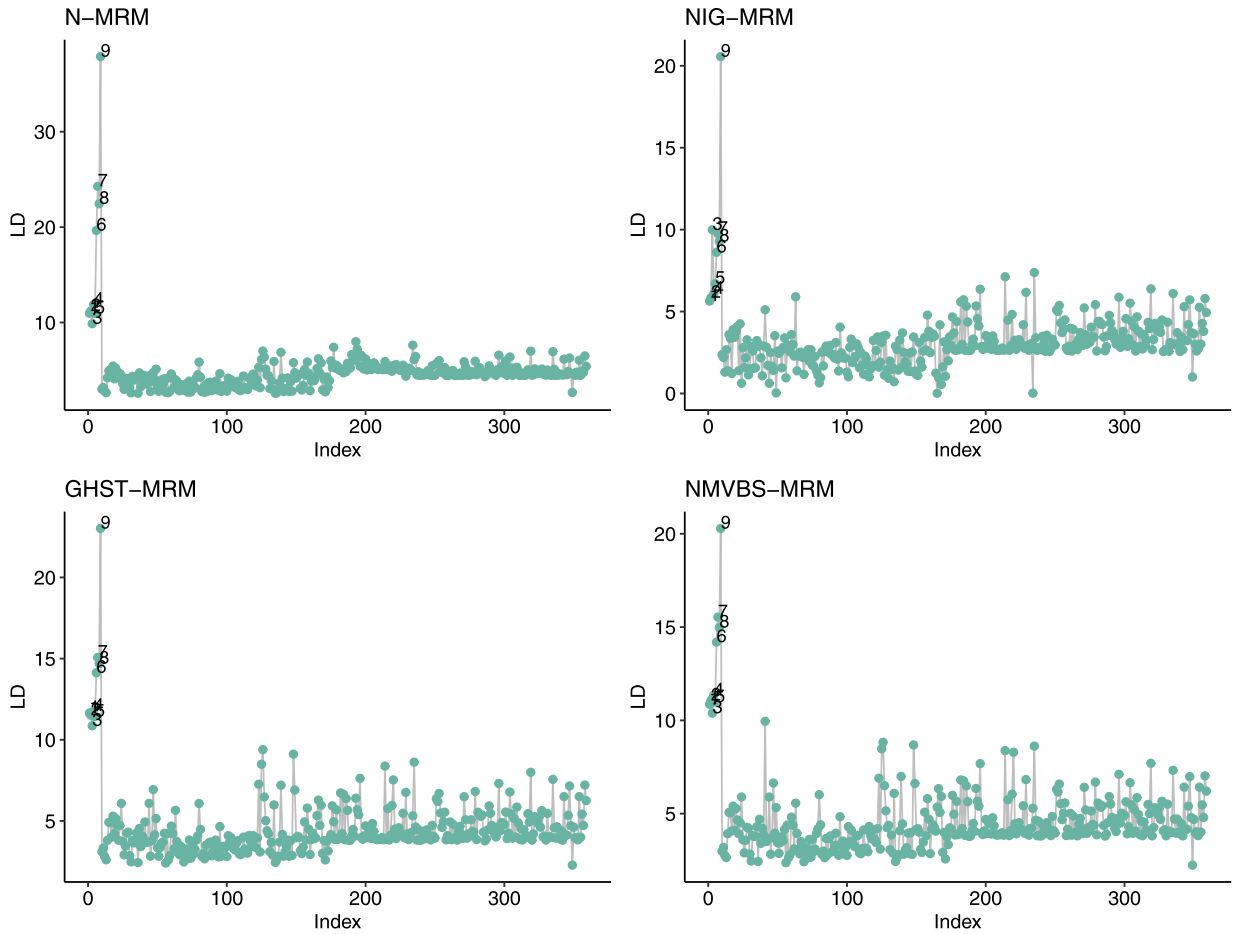


Fig. 4. Plot of LD measures for the *Pinus nigra* data fitted by four selected MRMs.

As for λ_i , we set $\lambda_1 = 2$, $\lambda_2 = 3$ and $\lambda_3 = 4$. Moreover, we set $\mathbf{x}_j = (1, x_{1j})^\top$, where x_{1j} is randomly drawn from $U(-1, 5)$. Fig. 6 shows the scatter plots of one simulated sample for each of parallel and concurrent cases, where different colors and digits represent the distinct groups of individuals.

In each replication, we randomly draw an observation, say $y_{(l)}$, and add 20 perturbations on it, namely $y_{(l)}^* = y_{(l)} - v$ with v varying from 1 to 20. Six members of NMVM-MRMs are fitted to the data with and without contamination. The resulting parameter estimates are denoted by $\hat{\theta}^{(v)}$ and $\hat{\theta}^{(0)}$, respectively. To assess the influence of a single outlier on the estimates, we calculate the mean magnitude of relative error (MMRE), defined as

$$\text{MMRE} = \frac{1}{12} \sum_{i=1}^3 \left\{ \left| \frac{\hat{\beta}_{0i}^{(v)} - \hat{\beta}_{0i}^{(0)}}{\hat{\beta}_{0i}^{(0)}} \right| + \left| \frac{\hat{\beta}_{1i}^{(v)} - \hat{\beta}_{1i}^{(0)}}{\hat{\beta}_{1i}^{(0)}} \right| + \left| \frac{\hat{\sigma}_i^{2(v)} - \hat{\sigma}_i^{2(0)}}{\hat{\sigma}_i^{2(0)}} \right| + \left| \frac{\hat{\lambda}_i^{(v)} - \hat{\lambda}_i^{(0)}}{\hat{\lambda}_i^{(0)}} \right| \right\}.$$

Fig. 7 displays the average MMRE as a function of contamination level v under the parallel and concurrent scenarios. It is readily seen that the influence of outliers in estimation increases as v is getting high. As expected, the GHST-MRM, NIG-MRM and NMVBS-MRM are generally less adversely affected under both scenarios. For models with fixed χ , namely NMVL-MRM and VG-MRM, it seems to lack enough resistance against outlying effects, at least for the results shown in this study.

6.3. Sensitivity of the EM estimates in presence of noise points

This simulation aims at checking the ability of our proposed model to cluster data in the presence of noise points. We simulate 200 samples of size $n = 500$ from (1) with $g = 3$. The error terms are assumed to be either the extended skew-normal (ESN; Capitanio et al. (2003)) or skew- t (ST; Azzalini and Capitanio (2003)) distributions with degree of freedoms $\nu_1 = 2$, $\nu_2 = 3$, and $\nu_3 = 5$. The covariates are set to $\mathbf{x}_j = (1, x_{1j})$ where x_{1j} is generated from $U(-1, 5)$. We also consider the parallel and concurrent scenarios with the same presumed parameters as Section 6.2. In order to verify the effect of

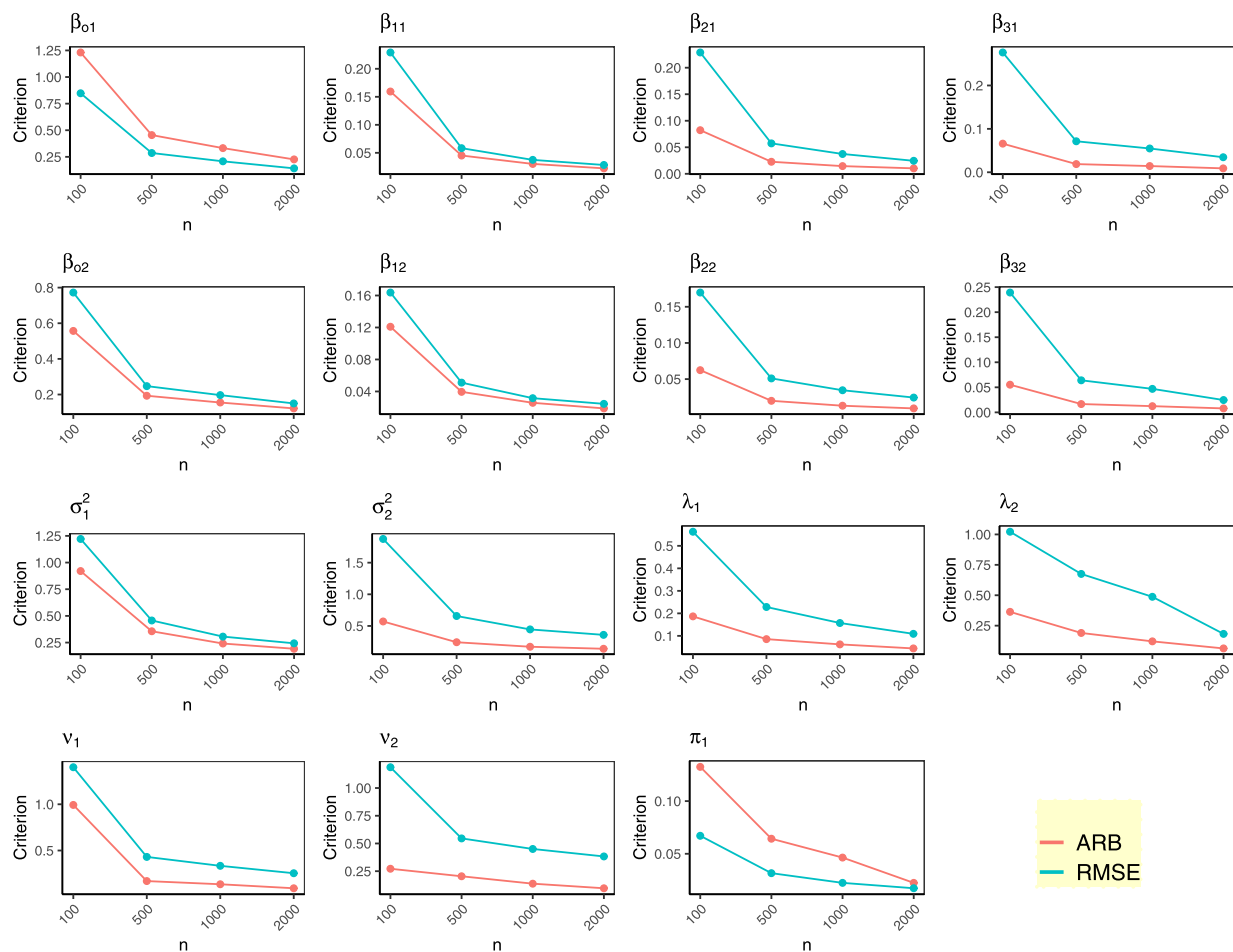


Fig. 5. Absolute relative bias (ARB) and the root mean squared error (RMSE) for the estimates of parameters across various sample sizes.

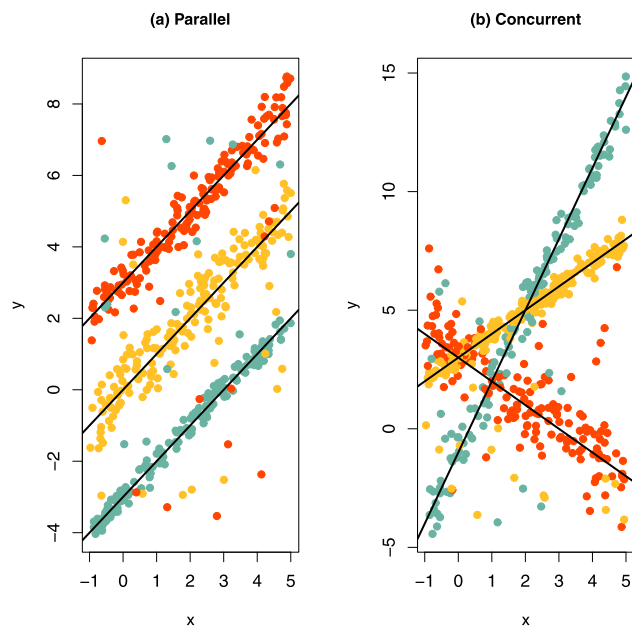


Fig. 6. Scatter plots of samples generated from the SN-MRM model with (a) parallel and (b) concurrent configurations.

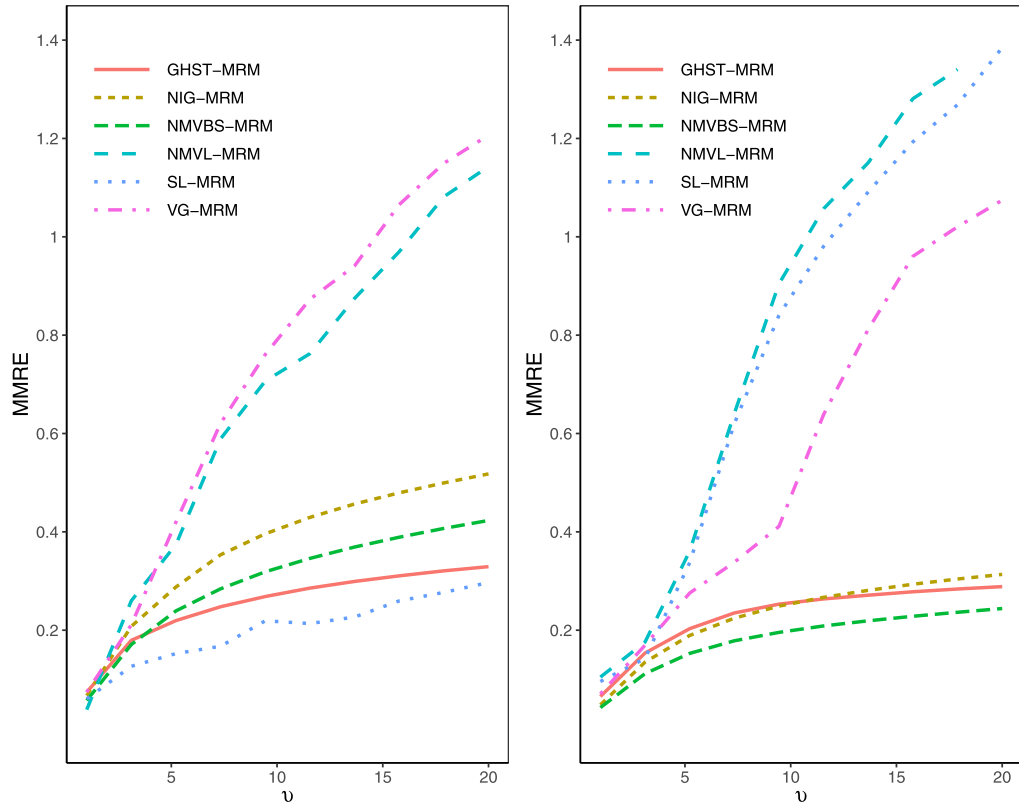


Fig. 7. Average MMRE as a function of contamination level (v) for the parallel (left panel) and concurrent (right panel) data-generating scenarios.

noise observations in model selection and clustering performance, we add 50 points generated from the $U(-4, 8)$ and assign them to one of the three classes randomly. Fig. 8 shows the scatter plots of the samples generated under each scenario.

In each replication of 200 trials, we fit six NMVM-MRMs with $g = 3$ and compute the resulting CAIC and ABIC scores. The adjusted Rand index (ARI; Hubert and Arabie, 1985) and the adjusted mutual information (AMI; Vinh et al., 2010) are used to measure the corresponding classification accuracy. Besides, the average CPU time (in second) for each scenario is recorded. As shown in Table 4, when the error terms are generated under the ESN (light-tailed) data-generation scenario, the NIG-MRM, GHST-MRM, and NMVBS-MRM give slightly better fitting performance than the other counterparts. Although the model selection and cluster validation indices for these MRMs are very close for 0% noise points, the NMVBS-MRM and NIG-MRM are generally favored because of their fast implementation in terms of less consumed CPU time. By adding 50 noise points, the results are somewhat similar for the concurrent configuration, while the GHST-MRM yields the best fitting and clustering performance for the parallel data generation. On the contrary, when the error terms are generated under the ST (heavy-tailed) scenario, the NIG-MRM and NMVBS-MRM outperform the other models for the uncontaminated (non-noisy) and contaminated (noisy) datasets, respectively.

6.4. Comparison to a competitive approach

In this simulation experiment, we compare the performance of our proposed model to the class of scale-mixture of skew-normal MRM (SMSN-MRM) introduced by Zeller et al. (2016). As an alternative skew extension of the classical MRM (1), the SMSN-MRM can be formulated as

$$\epsilon_j | (W_j = w_j, Z_{ij} = 1) \sim SN(0, w_j^{-1} \sigma_i^2, \lambda_i), \quad W_j \sim h(w_j; \mathbf{v}_i).$$

Give different mixing distributions for W_j , the SMSN-MRM includes the skew- t , skew-slash, and skew-contaminated normal MRMs as special cases, denoted by ST-MRM, SSL-MRM, and SCN-MRM, respectively. In the literature, there is no evidence to compare the practical performance between the NMVM and SMSN models. Generally, the users should consider both models in real analysis and decide which one judiciously describes data.

We generate 100 MC samples of size $n = 500$ from a 3-component MRM (1) with $\pi_1 = 0.3$ and $\pi_2 = 0.5$, where the remaining true parameters are specified as

$$\beta_1 = (-1, -1, -4, -3, -2)^\top, \quad \beta_2 = (1, 1, 4, 3, 2)^\top, \quad \beta_3 = (-1, 1, -4, 3, 2)^\top, \quad \text{and} \quad \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = 1.$$

Table 4

Simulation results for assessing model selection and clustering performance of the six sub-models of NMVM-MRM when the data are generated by the assumption of ESN and ST distributions on the error terms with and without adding noise points under parallel and concurrent configurations.

Error Generator	noise	Measure	SL-MRM		VG-MRM		NIG-MRM	
			parallel	concurrent	parallel	concurrent	parallel	concurrent
ESN	0	CAIC	1583.106	1671.222	1556.873	1649.150	1550.144	1642.714
		ABIC	1596.808	1684.924	1575.931	1668.207	1566.563	1659.133
		ARI	0.998	0.732	0.999	0.738	0.999	0.738
		AMI	0.994	0.655	0.994	0.662	0.994	0.663
		CPU time	0.670	0.436	10.860	12.809	2.233	2.762
	50	CAIC	2033.764	2068.521	2055.654	2067.903	1992.668	2055.842
		ABIC	2048.876	2083.633	2077.776	2088.025	2010.821	2073.995
		ARI	0.821	0.599	0.805	0.600	0.821	0.602
		AMI	0.756	0.520	0.739	0.520	0.751	0.523
		CPU time	0.837	0.507	17.825	21.912	2.303	2.752
ST	0	CAIC	1467.483	1593.508	1485.109	1604.343	1456.930	1575.560
		ABIC	1481.185	1607.210	1504.166	1623.400	1473.349	1591.980
		ARI	0.976	0.768	0.965	0.754	0.980	0.774
		AMI	0.961	0.691	0.941	0.674	0.966	0.697
		CPU.T	1.105	1.182	13.364	13.748	1.161	1.942
	50	CAIC	2371.044	2024.224	2401.626	2045.151	2291.921	2002.673
		ABIC	2387.437	2039.335	2424.620	2066.273	2311.647	2020.826
		ARI	0.640	0.620	0.648	0.614	0.647	0.622
		AMI	0.570	0.542	0.576	0.533	0.575	0.544
		CPU time	0.826	0.945	18.263	18.692	1.496	2.064
Error Generator	noise	Measure	GHST-MRM		NMVBS-MRM		NMVL-MRM	
			parallel	concurrent	parallel	concurrent	parallel	concurrent
ESN	0	CAIC	1551.413	1643.807	1550.125	1642.697	1577.978	1667.227
		ABIC	1567.832	1660.226	1566.545	1659.116	1594.397	1683.646
		ARI	0.998	0.739	0.999	0.740	0.998	0.734
		AMI	0.994	0.664	0.994	0.665	0.994	0.657
		CPU time	8.305	9.067	2.311	2.850	2.151	2.878
	50	CAIC	1987.678	2057.663	2012.507	2055.679	2050.823	2072.3493
		ABIC	2005.831	2075.816	2030.660	2073.832	2068.976	2090.503
		ARI	0.826	0.602	0.820	0.605	0.821	0.600
		AMI	0.760	0.523	0.751	0.527	0.755	0.520
		CPU time	14.045	15.206	2.541	2.907	2.247	2.889
ST	0	CAIC	1459.418	1577.723	1459.381	1579.832	1474.203	1600.717
		ABIC	1475.838	1594.143	1475.800	1596.25	1490.623	1617.137
		ARI	0.980	0.768	0.980	0.771	0.976	0.769
		AMI	0.965	0.691	0.965	0.694	0.961	0.691
		CPU time	12.453	13.061	1.329	2.143	1.989	2.356
	50	CAIC	2315.020	2011.547	2277.523	1999.634	2390.085	2033.493
		ABIC	2334.745	2029.700	2297.249	2017.787	2409.811	2051.646
		ARI	0.644	0.620	0.649	0.623	0.638	0.620
		AMI	0.571	0.542	0.578	0.545	0.569	0.541
		CPU time	15.927	18.798	1.623	2.334	2.198	3.331

Moreover, the covariates are set to $\mathbf{x}_j = (1, x_{1j}, x_{2j}, x_{3j}, x_{4j})^\top$, where x_{kj} ($k = 1, \dots, 4$) are randomly drawn from $U(-1, 5)$, $U(-1, 1)$, $U(0, 1)$, and $U(-2, 2)$ distributions, respectively. To make fair comparisons, the errors are generated from the mean-variance mixture of skew-normal (MVM-SN) distributions considered by Arslan (2015). Specifically, the errors take the form of $\epsilon_{ij} = W\lambda_i + \sqrt{W}Z_1$, where $Z_1 \sim SN(0, \sigma_1^2, 1)$, and independently $W \sim GIG(-\nu_i/2, \nu_i, 0)$ with $\lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 4, \nu_1 = 2, \nu_2 = 3, \nu_3 = 5$. In this setting, the simulated data are neither from NMVM nor SMSN class of MRMs, but allows for capturing the main characteristics of the two competitive models.

In each replication, three NMVM-MRMs (GHST, NIG and NMVBS) and three SMSN-MRM (ST, SSL, SCN) are considered to fit the simulate dataset with MVM-SN errors. The BIC value and miss-classifications error rate (MCR) are recorded. Note that a smaller value of the two criteria indicates a better model calibration. Fig. 9 shows the heat-map pairwise comparison of six models to determine which model is more suited to the simulated data. The values in the r th row ($r = 1, \dots, 6$) of the heat-map plot represent the number of times that model outperforms the others in the s th column ($s \neq r$). The alongside bar diagrams show the cumulative pairwise comparison frequencies for each row. As can be seen, the NMVBS-MRM offers the best overall fit to the data, while the GHST-MRM yields the greatest gross clustering performance. Comparing the overall total frequencies for two classes of MRMs in Fig. 9, our proposed NMVBS-MRM is shown to be more suited for this data generation than SMSN-MRMs.

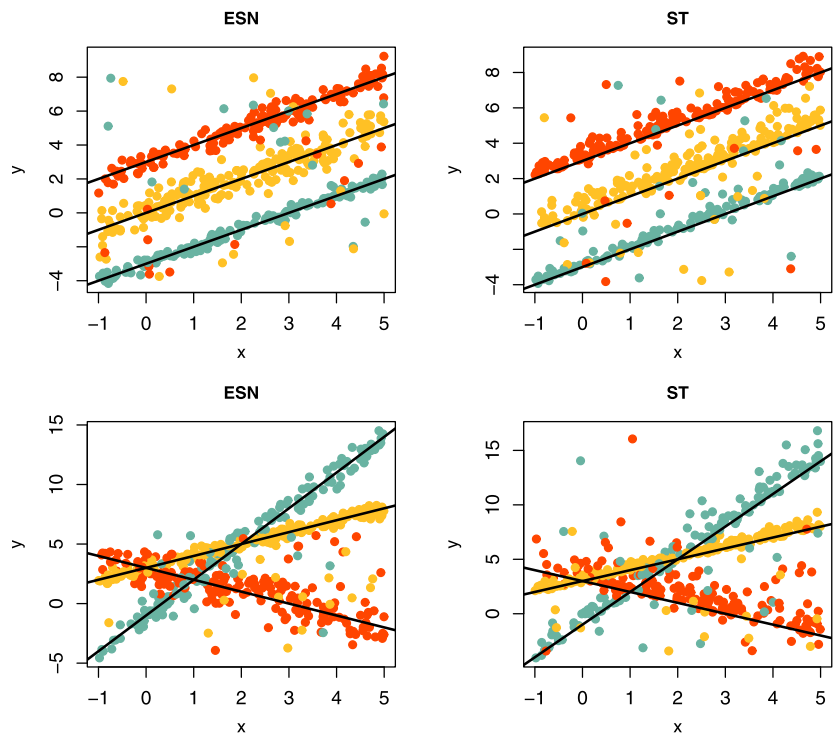


Fig. 8. Scatter plots of samples generated under parallel (upper panel) and concurrent (down panel) scenarios with ESN and ST errors.

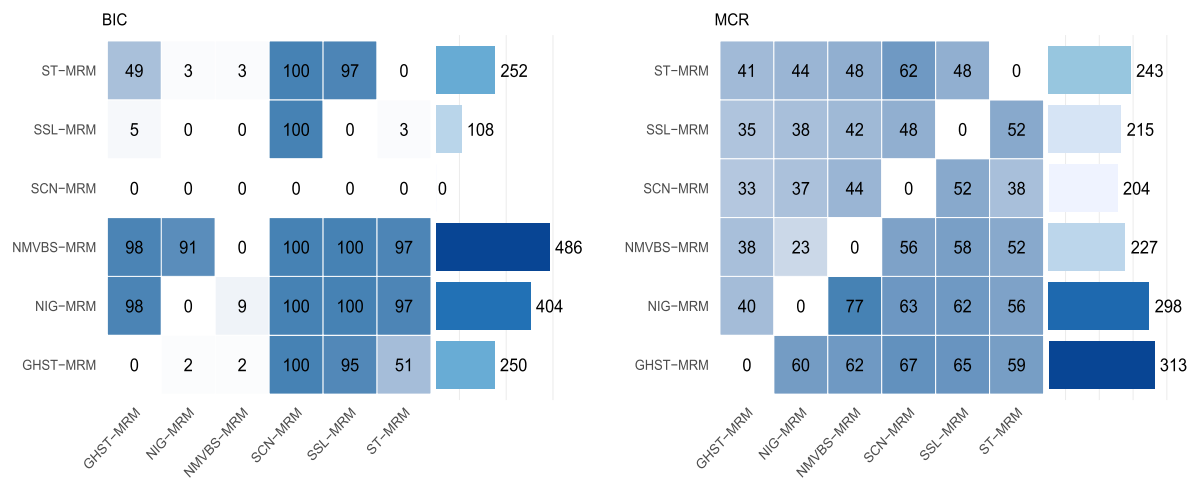


Fig. 9. Pairwise comparisons of NMVM-MRM and SMSNMRM in terms of the BIC and MCR scores.

7. Concluding remarks

We have introduced a robust extension of MRMs based on the class of NMVM distributions, called the NMVM-MRM, which is a novel tool alternative to existing methods for investigating the complex relationship of random phenomena arising from several heterogeneous sub-populations. A computationally efficient ECME algorithm is developed to compute the ML estimates of parameters under a convenient hierarchical representation of the model. The issue of approximating asymptotic standard errors of parameter estimates is also addressed through the information matrix-based approach. Numerical results from real-data applications have revealed that the proposed model can offer strong resistance against the influence of outliers and noise points. Additionally, simulation experiments show the capacity of our proposed computational techniques on the provision of desirable asymptotic properties of ML estimates. The computer programs coded in the R language for implementation of our methods are available at <https://github.com/MerNaderi/NMVM-MRM.git>.

There are some possible variants of the current approach that deserve further exploration. One possible direction is to consider the NMVM distributions for the random variates of the mixture of linear experts (MoE) model (Jacobs et al., 1991) with possibly censored data (Mirfarah et al., 2021). It is also worthy to develop a fully Bayesian approach to the MoE model based on the rich class of NMVM distributions. Finite mixtures of linear mixed-effects (FM-LME) models (Verbeke and Lesaffre, 1996) enable the analysts to approximate the underlying distribution of data with observations measured at arbitrary sets of irregular occasions originating possibly from heterogeneous groups described by different component fixed-effects and variance parameters. Robust extensions of the FM-LME model using the multivariate Student's t distribution were considered by Bai et al. (2016) and Yang et al. (2020) for single-outcome repeated measures and by Wang (2017) and Wang (2019) for multi-outcome longitudinal data. Another promising avenue for future work is to pursue a flexible expansion of the (multivariate) FM-LME model with NMVM component random effects and errors. The methodology presented in this paper can also be extended by considering the family of skew-normal generalized hyperbolic distributions (Vilca et al., 2014) which can provide more flexibility in modeling data with strong skewness and heavy-tailed features.

The linear cluster-weighted model (CWM) introduced by Gershensfeld (1997) provides a different mixture approach to modeling the joint probability of a response variable and a set of explanatory variables. The linear CWM model and its robust variants are very useful for multivariate regression analysis with random covariates in the presence of unobserved group heterogeneity, see, e.g., Ingrassia et al. (2014) and Ingrassia et al. (2015) for a more detailed account of their potential capabilities. The general setting of CWM is to assume that the response variable y_j and the vector of covariates $\mathbf{x}_j$, coming from a population Ω assume to be partitioned into g disjoint groups $\Omega_1, \dots, \Omega_g$, have a joint distribution $f(y_j, \mathbf{x}_j) = \sum_{i=1}^g \pi_i f(\mathbf{x}_j | \Omega_i) f(y_j | \mathbf{x}_j, \Omega_i)$. Ingrassia et al. (2012) clarified the relationship between the linear CWM and MRM and proved that the two model may lead to the same pdf under suitable assumptions. Inspired by one referee's suggestion, we are interested in developing a new robust CWM model based on the class of NMVM distributions and hope some interesting findings can be reported in a follow-up paper.

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Appendix A. Model formulation of the NMVM distribution

A random variable X is said to follow a NMVM distribution, $X \sim \text{NMVM}(\mu, \lambda, \sigma^2, \mathbf{v})$, if it has the stochastic representation:

$$X = \mu + W\lambda + \sqrt{W}Z, \quad W \perp Z, \quad (\text{A.1})$$

where $\mu \in \mathbb{R}$, $\lambda \in \mathbb{R}$, $Z \sim N(0, \sigma^2)$, W is an arbitrary univariate positive random variable with pdf $h(\cdot; \mathbf{v})$ indexed by a possibly parameter vector $\mathbf{v}$, and ' $\perp$ ' indicates independence. The convenient hierarchical representation of the NMVM distribution is given by (3). Accordingly, the probability density function (pdf) of X can be derived through

$$f_{\text{NMVM}}(x; \mu, \lambda, \sigma^2, \mathbf{v}) = \int_0^\infty \phi(x; \mu + w\lambda, w\sigma^2) h(w; \mathbf{v}) dw. \quad (\text{A.2})$$

An attractive sub-class of the NMVM family can be constructed by assuming the GIG distribution on W , denoted by $W \sim \text{GIG}(\kappa, \chi, \psi)$, with pdf

$$h_{\text{GIG}}(w; \kappa, \chi, \psi) = \left(\frac{\psi}{\chi}\right)^{\kappa/2} \frac{w^{\kappa-1}}{2K_\kappa(\sqrt{\psi\chi})} \exp\left\{-\frac{1}{2}(w^{-1}\chi + w\psi)\right\}, \quad w > 0,$$

where

$$K_\kappa(x) = \frac{1}{2} \int_0^\infty y^{\kappa-1} \exp\left(-\frac{x}{2}\left(y + \frac{1}{y}\right)\right) dy$$

is the modified Bessel function of the third kind with $x > 0$, the index $\kappa \in \mathbb{R}$, and two parameters χ and ψ satisfying the conditions: (i) $\chi \geq 0, \psi > 0$ if $\kappa > 0$; (ii) $\chi > 0, \psi \geq 0$ if $\kappa < 0$; and (iii) $\chi > 0, \psi > 0$ if $\kappa = 0$. Denoting by $R_{(\kappa,a)}(c) = K_{\kappa+a}(c)/K_\kappa(c)$, we have for $r = \pm 1$:

$$E(W^r) = \left(\frac{\chi}{\psi}\right)^{r/2} R_{(\kappa,r)}(\sqrt{\chi\psi}) \quad \text{and} \quad E(\log W) = \log(\sqrt{\chi/\psi}) + \frac{1}{K_\kappa(\chi\psi)} \frac{\partial K_\kappa(\chi\psi)}{\partial \kappa}. \quad (\text{A.3})$$

Following representation (A.1) coupled with the assumption of $W \sim \text{GIG}(\kappa, \chi, \psi)$, X is said to have the GH distribution with the pdf

$$f_{\text{GH}}(x; \mu, \lambda, \sigma^2, \kappa, \chi, \psi) = \frac{\left(\frac{\psi}{\chi}\right)^{\kappa/2} \left(\psi + \frac{\lambda^2}{\sigma^2}\right)^{0.5-\kappa} K_{\kappa-0.5} \left(\sqrt{(\psi + \lambda^2/\sigma^2)(\chi + \delta(x; \mu, \sigma^2))} \right)}{\sqrt{2\pi\sigma^2} K_{\kappa}(\sqrt{\psi\chi}) \left(\sqrt{(\psi + \lambda^2/\sigma^2)(\chi + \delta(x; \mu, \sigma^2))} \right)^{0.5-\kappa}} \exp \left\{ \frac{(x - \mu)\lambda}{\sigma^2} \right\},$$

where $\delta(x, \mu, \sigma^2) = (x - \mu)^2/\sigma^2$.

Appendix B. Some particular members of the NMVM family

Hereby, we discuss some particular members of the NMVM distributions and present the conditional expectations required for the proposed ECME algorithm.

- **The generalized hyperbolic skew- t distribution:** Suppose that the mixing variable W in (3) has an inverse gamma distribution, denoted by $W \sim \text{Invgamma}(v/2, v/2)$, with mean $E(W) = v/(v-2)$ and variance $\text{Var}(W) = (v-2)^{-2}(v-4)^{-1}2v^2$. In this case, the random variable X has the GHST distribution, which is a limiting case of the GH distribution by taking $\kappa = -v/2$, $\chi = v$ and $\psi = 0$. The GHST distribution has the pdf

$$f_{\text{GHST}}(x; \mu, \lambda, \sigma^2, v) = \left(\frac{\lambda^2/\sigma^2}{v + \delta(x, \mu, \sigma^2)} \right)^{(v+1)/4} \frac{v^{v/2} K_{\frac{v+1}{2}} \left(\sqrt{\lambda^2(v + \delta(x, \mu, \sigma^2))/\sigma^2} \right)}{\sqrt{2\pi\sigma^2} \Gamma(v/2) 2^{v/2-1} \exp \{(\mu - x)\lambda/\sigma^2\}},$$

where $\delta(x, \mu, \sigma^2) = (x - \mu)^2/\sigma^2$. By Bayes' rule, we can deduce that $W | (X = x) \sim \text{GIG}(\kappa_{ST}^*, \chi_{ST}^*, \psi_{ST}^*)$, where $\kappa_{ST}^* = -(v+1)/2$, $\chi_{ST}^* = \delta(x, \mu, \sigma^2) + v$, $\psi_{ST}^* = \lambda^2/\sigma^2$. Using (A.3), the r th moment of W given $X = x$ can be easily obtained.

- **The normal inverse Gaussian distribution:** The NIG distribution is a special case of the GH distribution by taking $\kappa = -0.5$, $\chi = 1$, and $\psi = v^2$. Using $K_{0.5}(x) = K_{-0.5}(x) = \sqrt{\pi/2} x^{-0.5} \exp\{-x\}$, the pdf of the NIG distribution is given by

$$f_{\text{NIG}}(x; \mu, \lambda, \sigma^2, v) = \sqrt{\frac{\psi_{\text{NIG}}^*}{\pi^2 \sigma^2 \chi_{\text{NIG}}^*}} K_1 \left(\sqrt{\chi_{\text{NIG}}^* \psi_{\text{NIG}}^*} \right) \exp \left\{ \frac{(x - \mu)\lambda}{\sigma^2} + v \right\},$$

where $\chi_{\text{NIG}}^* = 1 + \delta(x, \mu, \sigma^2)$, $\psi_{\text{NIG}}^* = v^2 + (\lambda/\sigma)^2$. Thus, it can be deduced that $W | (X = x) \sim \text{GIG}(-1, \chi_{\text{NIG}}^*, \psi_{\text{NIG}}^*)$, and so the r th moment of W given $X = x$ can be obtained by (A.3).

- **The variance-gamma distribution:** As another reduced version of the GH distribution, the VG distribution is obtained when $\chi = 0$ and $\kappa > 0$. The pdf of the VG distribution is given by

$$f_{\text{VG}}(x; \mu, \lambda, \sigma^2, \kappa, \psi) = \frac{\psi^{\kappa} K_{\kappa-0.5} \left(\sqrt{\psi_{\text{VG}}^* \chi_{\text{VG}}^*} \right)}{2^{\kappa-1} \sqrt{2\pi\sigma^2} \Gamma(\kappa) \left(\sqrt{\chi_{\text{VG}}^* / \psi_{\text{VG}}^*} \right)^{0.5-\kappa}} \exp \left\{ \frac{(x - \mu)\lambda}{\sigma^2} \right\},$$

where $\chi_{\text{VG}}^* = \delta(x, \mu, \sigma^2)$ and $\psi_{\text{VG}}^* = \psi + (\lambda/\sigma)^2$. Similarly, it suffices to show that $W | (X = x) \sim \text{GIG}(\kappa - 0.5, \chi_{\text{VG}}^*, \psi_{\text{VG}}^*)$. The r th moment of W given $X = x$ for the VG distribution can be obtained by (A.3) accordingly.

- **The skew Laplace distribution:** The SL distribution is created by assuming that W in (3) has an exponential distribution with mean 0.5, denoted by $W \sim \text{Exp}(0.5)$. In this case, the pdf of X is

$$f_{\text{SL}}(x; \mu, \sigma^2, \lambda) = \frac{1}{2\tau\sigma} \exp \left\{ -\tau \frac{|x - \mu|}{\sigma} + \frac{\lambda(x - \mu)}{\sigma^2} \right\},$$

where $\tau = \sqrt{1 + (\lambda/\sigma)^2}$. By Bayes' rule, it is easy to verify that $W | (X = x) \sim \text{GIG}(0.5, \chi_{\text{SL}}^*, \psi_{\text{SL}}^*)$, where $\chi_{\text{SL}}^* = \delta(x, \mu, \sigma^2)$ and $\psi_{\text{SL}}^* = \tau^2$. Consequently, the r th moment of W given $X = x$ can be obtained by invoking (A.3).

- **The normal mean-variance Birnbaum-Saunders distribution:** Let W be a positive random variable following the Birnbaum-Saunders (BS) distribution (Birnbaum and Saunders, 1969) with the shape and scale parameters α and β , denoted by $W \sim \text{BS}(\alpha, \beta)$. Following Desmond (1986), the pdf of $W \sim \text{BS}(\alpha, \beta)$ can be expressed as a mixture of two GIG distributions, given by

$$h_{\text{BS}}(w; \alpha, \beta) = \frac{1}{2} f_{\text{GIG}} \left(w; \frac{1}{2}, \frac{\beta}{\alpha^2}, \frac{1}{\beta\alpha^2} \right) + \frac{1}{2} f_{\text{GIG}} \left(w; -\frac{1}{2}, \frac{\beta}{\alpha^2}, \frac{1}{\beta\alpha^2} \right).$$

The NMVBS distribution introduced by Pourmoussa et al. (2015) is established by assuming $W \sim \text{BS}(\alpha, 1)$ as a mixing variable. In this setup, the pdf of X is

$$f_{\text{NMVBS}}(x; \mu, \lambda, \sigma^2, \alpha) = \frac{1}{2} f_{\text{GH}}\left(x; \mu, \lambda, \sigma^2, \frac{1}{2}, \frac{1}{\alpha^2}, \frac{1}{\alpha^2}\right) + \frac{1}{2} f_{\text{GH}}\left(x; \mu, \lambda, \sigma^2, -\frac{1}{2}, \frac{1}{\alpha^2}, \frac{1}{\alpha^2}\right).$$

By Theorem 3.2 in Naderi et al. (2017), for $r = \pm 1, \pm 2, \dots$, we have

$$E(W^r | X = x) = \left(\frac{\chi_{\text{BS}}^*}{\psi_{\text{BS}}^*}\right)^{r/2} \left\{ p(x) R_{(0,r)}\left(\sqrt{\chi_{\text{BS}}^* \psi_{\text{BS}}^*}\right) + (1 - p(x)) R_{(-1,r)}\left(\sqrt{\chi_{\text{BS}}^* \psi_{\text{BS}}^*}\right) \right\},$$

where $\chi_{\text{BS}}^* = \delta(x, \mu, \sigma^2) + \alpha^{-2}$, $\psi_{\text{BS}}^* = (\lambda/\sigma)^2 + \alpha^{-2}$, and

$$p(x) = \frac{f_{\text{GH}}(x; \mu, \lambda, \sigma^2, 0.5, \alpha^{-2}, \alpha^{-2})}{f_{\text{GH}}(x; \mu, \lambda, \sigma^2, 0.5, \alpha^{-2}, \alpha^{-2}) + f_{\text{GH}}(x; \mu, \lambda, \sigma^2, -0.5, \alpha^{-2}, \alpha^{-2})}.$$

- **The normal mean-variance Lindley distribution:** The NMVL distribution introduced by Naderi et al. (2018) is fabricated by assuming $W \sim \text{Lindley}(\alpha)$, where $\text{Lindley}(\alpha)$ denotes the Lindley distribution (Lindley, 1958) with pdf $h_{\text{Lindley}}(w | \alpha) \propto \alpha^2(1+w) \exp[-\alpha w]$ for $\alpha > 0$ and $w > 0$. The pdf of X can therefore be written as

$$f_{\text{NMVL}}(x; \mu, \lambda, \sigma^2, \alpha) = \frac{\alpha}{\alpha + 1} f_{\text{GH}}(x; \mu, \lambda, \sigma^2, 1, 0, 2\alpha) + \frac{1}{\alpha + 1} f_{\text{GH}}(x; \mu, \lambda, \sigma^2, 2, 0, 2\alpha).$$

By Proposition 1 of Naderi et al. (2018), for $r = \pm 1, \pm 2, \dots$, we get

$$E(W^r | X = x) = \left(\frac{\chi_{\text{LI}}^*}{\psi_{\text{LI}}^*}\right)^{r/2} \left\{ p(x) R_{(0.5,r)}\left(\sqrt{\chi_{\text{LI}}^* \psi_{\text{LI}}^*}\right) + (1 - p(x)) R_{(1.5,r)}\left(\sqrt{\chi_{\text{LI}}^* \psi_{\text{LI}}^*}\right) \right\},$$

where $\chi_{\text{LI}}^* = \delta(x, \mu, \sigma^2)$, $\psi_{\text{LI}}^* = (\lambda/\sigma)^2 + 2\alpha$, and

$$p(x) = \frac{\alpha f_{\text{GH}}(x; \mu, \lambda, \sigma^2, 1, 0, 2\alpha)}{\alpha f_{\text{GH}}(x; \mu, \lambda, \sigma^2, 1, 0, 2\alpha) + f_{\text{GH}}(x; \mu, \lambda, \sigma^2, 2, 0, 2\alpha)}.$$

An obvious identifiability problem occurs because $f_{\text{GH}}(x; \mu, \lambda, \sigma^2, \kappa, \chi, \psi) = f_{\text{GH}}(x; \mu, a\lambda, a\sigma^2, \kappa, \chi/a, a\psi)$ for any constant $a \in \mathbb{R}^+$. To address this issue, Browne and McNicholas (2015) suggested an alternative parameterization for the GIG distribution. Browne and McNicholas (2015) set $\chi = \omega\eta$ and $\psi = \omega/\eta$, and showed that the finite mixture of GH distributions under this setting is identifiable. Following the notion of McNeil et al. (2005) and the reparameterization scheme of Browne and McNicholas (2015), the special cases of the GH distribution considered in the above are all identifiable.

References

- Aitken, A.C., 1927. On Bernoulli's numerical solution of algebraic equations. *Proc. R. Soc. Edinb.* 46, 289–305.
- Aitkin, M., Wilson, G.T., 1980. Mixture models, outliers, and the EM algorithm. *Technometrics* 22, 325–331.
- Akaike, H., 1998. Information theory and an extension of the maximum likelihood principle. In: *Selected Papers of Hirotugu Akaike*. Springer, pp. 199–213.
- Arslan, O., 2015. Variance-mean mixture of the multivariate skew normal distribution. *Stat. Pap.* 56, 353–378.
- Azzalini, A., Capitanio, A., 2003. Distributions generated by perturbation of symmetry with emphasis on a multivariate skew t -distribution. *J. R. Stat. Soc. B* 65, 367–389.
- Bai, X., Chen, K., Yao, W., 2016. Mixture of linear mixed models using multivariate t distribution. *J. Stat. Comput. Simul.* 86, 771–787.
- Barndorff-Nielsen, O., Halgreen, C., 1977. Infinite divisibility of the hyperbolic and generalized inverse Gaussian distributions. *Z. Wahrscheinlichkeitstheor. Verw. Geb.* 38, 309–311.
- Basford, K., Greenway, D., McLachlan, G., Peel, D., 1997. Standard errors of fitted means under normal mixture models. *Comput. Stat.* 12, 1–17.
- Bedrick, E.J., Tsai, C.L., 1994. Model selection for multivariate regression in small samples. *Biometrics*, 226–231.
- Benaglia, T., Chauveau, D., Hunter, D.R., 2009. An EM-like algorithm for semi- and nonparametric estimation in multivariate mixtures. *J. Comput. Graph. Stat.* 18, 505–526.
- Benites, L., Maehara, R., Lachos, V.H., Bolfarine, H., 2019. Linear regression models using finite mixtures of skew heavy-tailed distributions. *Chil. J. Stat.* 10, 319–327.
- Birnbaum, Z.W., Saunders, S.C., 1969. A new family of life distributions. *J. Appl. Probab.*, 319–327.
- Browne, R.P., McNicholas, P.D., 2015. A mixture of generalized hyperbolic distributions. *Can. J. Stat.* 43, 176–198.
- Capitanio, A., Azzalini, A., Stanghellini, E., 2003. Graphical models for skew-normal variates. *Scand. J. Stat.* 30, 129–144.
- Cohen, E.A., 1984. Some effects of inharmonic partials on interval perception. *Music Percept.* 1, 323–349.
- Cook, R.D., Weisberg, S., 1982. *Residuals and Influence in Regression*. Chapman and Hall, New York.
- Dempster, A.P., Laird, N.M., Rubin, D.B., 1977. Maximum likelihood from incomplete data via the EM algorithm. *J. R. Stat. Soc., Ser. B, Stat. Methodol.* 39, 1–22.
- Desmond, A., 1986. On the relationship between two fatigue-life models. *IEEE Trans. Reliab.* 35, 167–169.
- Frühwirth-Schnatter, S., 2006. *Finite Mixture and Markov Switching Models*. Springer, New York.
- Galimberti, G., Soffritti, G., 2014. A multivariate linear regression analysis using finite mixtures of t distributions. *Comput. Stat. Data Anal.* 71, 138–150.
- García-Escudero, L., Gordaliza, A., Mayo-Iscar, A., Martín, R.S., 2010. Robust clusterwise linear regression through trimming. *Comput. Stat. Data Anal.* 54, 3057–3069.
- García-Escudero, L.A., Gordaliza, A., 1999. Robustness properties of k means and trimmed k means. *J. Am. Stat. Assoc.* 94, 956–969.
- Gershensfeld, N., 1997. Nonlinear inference and cluster-weighted modeling. *Ann. N.Y. Acad. Sci.* 808, 18–24.
- Goldfeld, S., Quandt, R., 1973. A Markov model for switching regression. *J. Econom.* 1, 3–15.

- Good, I.J., 1953. The population frequencies of species and the estimation of population parameters. *Biometrika* 40, 237–264.
- Hartigan, J.A., Wong, M.A., 1979. Algorithm AS 136: a k-means clustering algorithm. *J. R. Stat. Soc., Ser. C, Appl. Stat.* 28, 100–108.
- Hennig, C., 2000. Identifiability of models for clusterwise linear regression. *J. Classif.* 17, 273–296.
- Hubert, L., Arabie, P., 1985. Comparing partitions. *J. Classif.* 2, 193–218.
- Hunter, D.R., Young, D.S., 2012. Semiparametric mixtures of regressions. *J. Nonparametr. Stat.* 24, 19–38.
- Ingrassia, S., Minotti, S.C., Vittadini, G., 2012. Local statistical modeling via the cluster-weighted approach with elliptical distributions. *J. Classif.* 29, 363–401.
- Ingrassia, S., Minotti, S.C., Punzo, A., 2014. Model-based clustering via linear cluster-weighted models. *Comput. Stat. Data Anal.* 71, 159–182.
- Ingrassia, S., Punzo, A., Vittadini, G., Minotti, S.C., 2015. The generalized linear mixed cluster-weighted model. *J. Classif.* 32, 85–113.
- Jacobs, R.A., Jordan, M.I., Nowlan, S.J., Hinton, G.E., 1991. Adaptive mixtures of local experts. *Neural Comput.* 3, 79–87.
- Lindley, D., 1958. Fiducial distributions and Bayes theorem. *J. R. Stat. Soc., Ser. B, Stat. Methodol.* 20, 102–107.
- Liu, C., Rubin, D.B., 1994. The ECM algorithm: a simple extension of EM and ECM with faster monotone convergence. *Biometrika* 81, 633–648.
- Liu, M., Lin, T.I., 2014. A skew-normal mixture regression model. *Educ. Psychol. Meas.* 74, 139–162.
- Louis, T.A., 1982. Finding the observed information matrix when using the EM algorithm. *J. R. Stat. Soc., Ser. B, Stat. Methodol.* 44, 226–233.
- Mazza, A., Punzo, A., 2017. Mixtures of multivariate contaminated normal regression models. *Stat. Pap.* 61, 787–822.
- McNeil, A.J., Frey, R., Embrechts, P., 2005. *Quantitative Risk Management: Concepts, Techniques and Tools-Revised Edition*. Princeton University Press.
- Meng, X.L., Rubin, D.B., 1993. Maximum likelihood estimation via the ECM algorithm: a general framework. *Biometrika* 80, 267–278.
- Mirfarah, E., Naderi, M., Chen, D.G., 2021. Mixture of linear experts model for censored data: a novel approach with scale-mixture of normal distributions. *Comput. Stat. Data Anal.* 158, 107182.
- Naderi, M., Arabpour, A., Jamalizadeh, A., 2018. Multivariate normal mean-variance mixture distribution based on Lindley distribution. *Commun. Stat., Simul. Comput.* 47, 1179–1192.
- Naderi, M., Arabpour, A., Lin, T.I., Jamalizadeh, A., 2017. Nonlinear regression models based on the normal mean-variance mixture of Birnbaum–Saunders distribution. *J. Korean Stat. Soc.* 46, 476–485.
- Naderi, M., Hung, W.L., Lin, T.I., Jamalizadeh, A., 2019. A novel mixture model using the multivariate normal mean-variance mixture of Birnbaum–Saunders distributions and its application to extrasolar planets. *J. Multivar. Anal.* 171, 126–138.
- Pourmousa, R., Jamalizadeh, A., Rezapour, M., 2015. Multivariate normal mean-variance mixture distribution based on Birnbaum–Saunders distribution. *J. Stat. Comput. Simul.* 85, 2736–2749.
- Punzo, A., McNicholas, P.D., 2017. Robust clustering in regression analysis via the contaminated Gaussian cluster-weighted model. *J. Classif.* 34, 249–293.
- Quandt, R., 1972. A new approach to estimating switching regressions. *J. Am. Stat. Assoc.* 67, 306–310.
- Richardson, S., Green, P.J., 1997. On Bayesian analysis of mixtures with an unknown number of components (with discussion). *J. R. Stat. Soc., Ser. B, Stat. Methodol.* 59, 731–792.
- Rousseeuw, P.J., Leroy, A.M., 2005. *Robust Regression and Outlier Detection*. John Wiley & Sons.
- Schreuder, H.T., Hafley, W.L., 1977. A useful bivariate distribution for describing stand structure of tree heights and diameters. *Biometrics* 33, 471–478.
- Schwarz, G., 1978. Estimating the dimension of a model. *Ann. Stat.* 6, 461–464.
- Sclove, S.L., 1987. Application of model-selection criteria to some problems in multivariate analysis. *Psychometrika* 52, 333–343.
- Song, W., Yao, W., Xing, Y., 2014. Robust mixture regression model fitting by Laplace distribution. *Comput. Stat. Data Anal.* 71, 128–137.
- Späth, H., 1979. Algorithm 39. Clusterwise linear regression. *Computing* 22, 367–373.
- Stephens, M., 2000. Dealing with label switching in mixture models. *J. R. Stat. Soc., Ser. B, Stat. Methodol.* 62, 795–809.
- Tzortzis, G., Likas, A., 2014. The MinMax K-means clustering algorithm. *Lect. Notes Comput. Sci.* 47, 2505–2516.
- Verbeke, G., Lesaffre, E., 1996. A linear mixed-effects model with heterogeneity in the random-effects population. *J. Am. Stat. Assoc.* 91, 217–221.
- Viele, K., Tong, B., 2002. Modeling with mixtures of linear regressions. *Stat. Comput.* 12, 315–330.
- Vilca, F., Balakrishnan, N., Zeller, C.B., 2014. Multivariate skew-normal generalized hyperbolic distribution and its properties. *J. Multivar. Anal.* 128, 73–85.
- Vinh, N.X., Epps, J., Bailey, J., 2010. Information theoretic measures for clusterings comparison: variants, properties, normalization and correction for chance. *J. Mach. Learn. Res.* 11, 2837–2854.
- Wang, W.L., 2017. Mixture of multivariate- t linear mixed models for multi-outcome longitudinal data with heterogeneity. *Stat. Sin.* 27, 733–760.
- Wang, W.L., 2019. Mixture of multivariate t nonlinear mixed models for multiple longitudinal data with heterogeneity and missing values. *Test* 28, 196–222.
- Yang, Y.C., Lin, T.I., Castro, L.M., Wang, W.L., 2020. Extending finite mixtures of t linear mixed-effects models with concomitant covariates. *Comput. Stat. Data Anal.* 148, 106961.
- Yao, W., Wei, Y., Yu, C., 2014. Robust mixture regression using the t -distribution. *Comput. Stat. Data Anal.* 71, 116–127.
- Zeller, C.B., Cabral, C.R., Lachos, V.H., 2016. Robust mixture regression modeling based on scale mixtures of skew-normal distributions. *Test* 25, 375–396.